ChemStock: QR-Enabled Mobile Inventory with Photo Handover, Agent Loss Tracking, and Geo-Tagged Chain of Custody for A and Aimee Laboratories (Cospachem Products)

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

Inventory management is an essential daily operation for product-based businesses, directly affecting financial accuracy, customer satisfaction, and organizational efficiency (Safira, 2023). In the Philippine context, micro, small, and medium enterprises (MSMEs) represent 99.5% of all business establishments and employ 63.2% of the national workforce (Philippine Statistics Authority, 2023). Despite their economic predominance, many MSMEs continue to operate using manual, paper-based systems for inventory tracking, sales documentation, and receivables management. This reliance on traditional methods leads to human error, data loss, and wasted resources, ultimately holding back their growth within an increasingly digital marketplace (Azmi et al., 2024).

A and Aimee Laboratories, operating under the trade name Cospachem Products, exemplifies these systemic challenges. The enterprise engages in the distribution of beauty and liniment products through a distributed sales model targeting teachers in educational institutions and employees in municipal offices across over 130 branches nationwide, while remaining accessible to general consumers. The operational structure comprises two primary components: the office side, where managers handle administrative documentation and inventory oversight, and the field side, consisting of sales representatives and collectors deployed to assigned territories. Operationally, the Cagayan de Oro branch acts as the primary logistical hub managing field territories across the entire province of Misamis Oriental, while the Butuan branch exercises complete operational jurisdiction over distribution channels throughout the entire Caraga region. Sales representatives request stocks from managers, conduct direct sales to customers, and document transactions using handwritten receipts provided by the company. These receipts are remitted weekly to managers, while a separate collector team visits customers months later to pursue outstanding payments. This operational cycle culminates annually when personnel from the Manila head office conduct branch visits to audit stocks sold and identify inventory discrepancies.

To understand the current situation, a structured interview was conducted with the regional office manager of both Cagayan de Oro and Butuan branches - Ms. Josephine M. Palangan and Vice Manager Mr. Sanny J. Alaan, respectively, which were selected because both branches co-work with each other, and regardless of branch location, operational processes remain consistent across Cospachem's network. The interview data revealed several operational problems, as detailed below.

The scale of operations involved a total of 39 staff members across two branches. The Butuan Branch located at purok 2, Lumbocan, Butuan City, comprised 22 workers, including 15 sales representatives, 6 collectors, and 1 vice branch manager. The Cagayan de Oro Branch located at Blk 13 Lot 25, Malvar St. Regency II, Iponan, Cagayan de Oro City, had 17 workers, consisting of 12 sales representatives, 4 collectors, and 1 branch manager. In terms of transaction volume, each sales representative handled an average of over 20 stocks daily, more than 120 stocks during weekly peak periods, and upward of 600 stocks per monthly sales period.

Current receiving procedures rely entirely on paper-based documentation. Delivery verification is conducted against forms received via Messenger, which are either printed or viewed on mobile devices. Physical counting of incoming stocks requires approximately one to two hours per delivery, with deliveries occurring monthly though intervals may extend to three months between order approval and shipment. Quantity and product discrepancies occur regularly, identified only through manual itemized post-delivery checks.

Stock release procedures mirror receiving processes in their manual nature. Weekly releases average a minimum of 150 assorted bottles per sales representative, with three to six representatives typically operating per branch. The organization maintains three teams in Butuan, each with supervisors who relay reports to the branch manager on weekends, and one team in Cagayan de Oro operating under the same protocol. Tracking items after they leave the office is a major problem; inventory is released but not properly accounted for in the field.

While the company maintains a "First In, First Out" (FIFO) policy whereby earliest stocks are prioritized for release and returned items are positioned for subsequent distribution, inventory monitoring reveals that expiration is not the primary concern. Interview data indicates that losses from expiration are relatively rare due to FIFO implementation. The predominant challenge involves losses attributable to sales representatives, where stocks are either lost in the field or unreported as sales, effectively becoming financial losses to the organization.

According to data obtained from the Branch Manager of the Cagayan de Oro branch, the financial impact of stock losses is substantial. The branch experiences annual stock losses averaging over 100 units. As a matter of company policy, the branch manager is held accountable and must reimburse the head office for the exact cost of the lost inventory. Consequently, the average annual repayment made by the manager amounts to over 30,000 Philippine Pesos.

Full inventory checks are conducted immediately before new shipments arrive, requiring agents to surrender all stocks on hand to enable accurate itemization of remaining inventory. This process necessitates individual counting of each item, accounting for returns, losses, and sales through manual draft sheets documenting item names, quantities released, quantities sold, and quantities returned. Discrepancies between recorded and actual stock are typically discovered only during head office audits, meaning errors accumulate throughout the year before identification, at which point tracing the source becomes extremely difficult.

The current information and communications technology (ICT) infrastructure at Cospachem Products remains entirely manual within its branch operations, relying exclusively on pen-and-paper methods, as no dedicated ICT hardware or software systems have been implemented at the branch level. The majority of branches depend solely on paper records, with only a small minority utilizing Microsoft Excel in a limited capacity. Critically, no digital systems, spreadsheets, or databases are employed for inventory tracking across the branches. This absence of automated systems has led to persistent operational errors, including overlooked items and unrecorded transactions, particularly during periods of high transaction volume when multiple product returns are processed simultaneously. It should be noted, however, that this manual setup applies exclusively to the branches and not to the head office in Manila, which may operate under different infrastructural arrangements.

Significant accountability gaps exist due to physical separation between stock release and agent receipt confirmation. Stock releases often occur without face-to-face interaction, with items sent via collectors, creating opportunities for disputes regarding quantities received. Sales representatives occasionally deny receiving products, and no systematic documentation exists to verify who received what and when.

When asked about the biggest inventory problem requiring a solution, the client specifically identified: “Managing deliveries and losses. I want a standardized format that includes the list of items, release date, sales, returns, and a remarks section to indicate any losses. I also need it to automatically send this information to the Sales Representatives I assign it to." Weekly reports are the primary requirement, with users interacting with the system on a weekly basis. The client expressed willingness to conduct staff training before full implementation.

The interview data paints a clear picture of an enterprise constrained by manual processes that create information gaps, accountability issues, and financial losses. The distributed nature of Cospachem's sales force, combined with the time lag between sales and collections, amplifies these challenges. Sales Representatives operate independently in the field, generating handwritten documentation that must physically travel back to the office. Collectors pursue payments months later, often with incomplete information. Managers struggle to maintain real-time visibility into stock positions, leading to the "unavoidable" losses the client described.

This study aims to design and develop an automated inventory management system tailored specifically to the unique operational structure of Cospachem Products. Drawing directly from the client's expressed needs, the proposed system will digitise stock requests and inventory reconciliation. By replacing manual paper trails with a centralized digital platform accessible to Managers and Sales Representatives. The system intends to enhance data accuracy, provide inventory visibility, establish verifiable chain of custody through geolocation tracking, minimize losses, and streamline reporting for the Manila head office and various branches (Sukron et al., 2024). The client's enthusiastic responses to questions about QR code scanning, photo documentation, and automated tracking confirm that the proposed technological intervention aligns with both the company's needs and its readiness for digital transformation.

1.1.1 Pre-Implementation Technology Acceptance Feasibility Study

To assess organizational readiness and validate the conceptual Task-Technology Fit (TTF) of the proposed intervention before full-scale programmatic development, the researchers conducted a pre-implementation feasibility study using a high-fidelity interactive prototype developed in Figma. The Technology Acceptance Model (TAM) framework was administered to a pilot sample of 38 active branch workforce members, comprising 2 managers (Manager & Vice Manager), 27 sales representatives, and 9 collectors across the regional research locales.

The empirical results from this baseline assessment demonstrated exceptional preliminary user acceptance. The proposed Stock Loss Tracking mechanism recorded the highest user evaluation with a mean score of 4.43 out of 5.00, confirming its critical importance to the workforce in mitigating branch liabilities. Both the Photo Proof for Chain of Custody and GeoTag Tracking modules achieved identical mean scores of 4.32 out of 5.00, validating the workforce's confidence in the system's spatial and visual accountability safeguards prior to deployment.

The Discrepancy Alert framework and Digital Stock Monitoring dashboard recorded favorable mean scores of 4.13 and 4.08, respectively. Cumulatively, the overall mobile adoption intent reached a mean score of 4.37 out of 5.00 (an 87.4% positive adoption metric). These baseline metrics signal a strong corporate readiness and workforce willingness for digital transformation, directly justifying the development of the formal ChemStock system.

TAM Core Feature Module	Measured Metric (Mean Score / 5.00)	Positive Adoption Equivalence	Workforce Operational Priority
Digital Stock Monitoring Dashboard	4.08	81.6%	Favorable
Stock Loss Tracking	4.43	88.6%	Highest Priority
Discrepancy Alert	4.13	82.6%	Favorable
GeoTag Tracking	4.32	86.4%	High Priority
Photo Proof for Chain of Custody	4.32	86.4%	High Priority
Cumulative Mobile Adoption Intent	4.37	87.4%	Strong Readiness for Transformation

Table 1. Pre-Implementation TAM Evaluation Results

1.2 Statement of the Problem

1.2.1 General Problem

Cospachem currently relies on a manual, paper-based inventory management system that leads to recurring stock losses, inaccurate record-keeping, and accountability disputes with Sales Representatives. It also causes unverifiable chain of custody during remote stock handovers and delayed discovery of discrepancies, which only surface during annual head office audits, resulting in financial losses and operational inefficiencies that hinder the company's growth and profitability.

1.2.2 Specific Problem

A. **Late Notice of Stock Losses (Measurable Financial Impact)**: Under the current manual system, branch managers receive no timely notification of stock losses. Discrepancies remain undetected until annual audits, delaying awareness and response. Annual stock losses average over 100 stocks per branch, costing managers approximately 30,000 Philippine Pesos in reimbursements. Without an early warning mechanism, late notification precludes root cause investigation, allowing recurring financial losses to persist unaddressed.

B. **Absence of On-Demand Inventory Visibility:** Managers have no mechanism to remotely view current stock levels without physically being present at the office. This forces the manager to photograph paper records while in the field and manually reconcile them later, delaying decision-making and allowing errors to go undetected for extended periods. While a small number of branches use Microsoft Excel, no centralized or real-time system exists across the 130+ branch network.

C. **Accountability Gaps and Unverifiable Chain of Custody**: Stock releases frequently occur without face-to-face interaction, as items are sent via collectors. This leaves no systematic, verifiable record of who received what, when, and where. Sales Representatives occasionally deny receiving products, and with 39 staff handling hundreds of transactions weekly, the lack of geographic tracking creates a completely unverifiable chain of custody.

D. **Manual Recording Errors and Cumbersome Inventory Counting:** The exclusive use of pen and paper leads to overlooked items and unrecorded transactions, particularly during high-volume periods involving multiple simultaneous returns. A single sales representative averages 20+ stocks daily and 120+ stocks during weekly peak periods, all recorded manually. Full inventory checks require counting each item individually, manually documenting item names, quantities released, quantities sold, and quantities returned, a time-consuming process that is highly prone to human error.

E. **Lack of Photographic and Geographical Evidence for Physical Stock Handovers**: Beyond the geographical gaps, there is no photographic evidence or digital timestamps for any receiving or releasing activity. When stock arrives damaged or quantities are disputed, it is extremely difficult to resolve arguments with Sales Representatives regarding the exact physical condition of the stock at the precise time of transfer.

1.3 Objectives of the Study

1.3.1 General Objective

To design, develop, and implement ChemStock, a QR-enabled mobile inventory management system for Cospachem Products that digitizes manual record-keeping, ensures a verifiable chain of custody, and improves the efficiency and accuracy of inventory reporting.

1.3.2 Specific Objective

A. To **design** and **develop** an Android-based platform that brings together key inventory processes such as receiving, releasing, returns, and recording losses into a single, centralized, and easily searchable system.

B. To **integrate** tracking and automated reconciliation features, such as QR code-based stock identification, geotagging, photo documentation, timestamps and customizable reporting, to enhance accountability, monitor stock handovers among Sales Representatives, and identify stock discrepancies efficiently.

C. To **test** and **evaluate** the system's performance and effectiveness, utilizing the following methods: User Acceptance Testing (UAT), time-motion study log, error and discrepancy log analysis, system usability scale (SUS), and pre and post implementation checklist.

1.4 Significance of the Study

This study holds significance for multiple stakeholders, each of whom will benefit from the successful development and implementation of the proposed system.

To Cospachem Management. The system will provide managers with visibility into inventory levels, sales performance, and agent accountability. The integration of geolocation tracking enables verification of stock release and receipt locations, reducing disputes about whether deliveries reached their intended destinations. This visibility enables data-driven decision-making regarding stock allocation, reordering, profitability, and Sales Representative performance evaluation. The reduction in stock losses directly improves profitability, addressing the current annual loss reimbursement of approximately 30,000+ Pesos per branch, while automated reporting removes the stress of manually compiling reports for head office audits.

To Sales Representatives. Sales Representatives will benefit from clear digital records of stocks received and sold, reducing disputes with management about accountability. Geotagged receipt confirmation protects agents from false accusations of non-receipt by providing objective evidence of when and where they acknowledged delivery. The system provides protection by documenting when products were received and by whom, preventing unfair accusations of loss.

To the Manila Head Office. Annual audits become more efficient and transparent when the branch can generate accurate, up-to-date reports on demand. The head office gains confidence in branch reporting and can focus on strategic oversight rather than forensic accounting of missing stocks.

To Future Researchers. This study documents the development of an inventory management system tailored to a unique Philippine business model involving distributed sales forces, credit-based transactions, and delayed collections. Future researchers studying inventory systems for MSMEs or businesses with similar structures can use this study as a reference point for system design, requirements gathering, and implementation strategies.

To Academic Institution. This capstone project demonstrates the institution's commitment to producing graduates capable of addressing real-world industry problems through technology. It serves as evidence of the practical application of computer science, information technology, and business management principles taught within the curriculum.

To the MSME Sector. As a case study of digital transformation in a Philippine MSME, this project illustrates how affordable, tailored technology solutions can address operational challenges faced by small businesses. The lessons learned and methodologies applied can inform similar initiatives across the sector, contributing to broader economic development.

1.5 Scope and Limitations

This study focuses on the design, development, and implementation of an automated inventory management system specifically for Cospachem Products' Cagayan de Oro and Butuan Branches (total 39 staff), with the understanding that operational processes are consistent across all 130+ branches nationwide, making the solution potentially scalable. The geographical scope of the implementation framework is strictly delimited to the distinct regional territories handled by these hubs: the Cagayan de Oro branch scope encompasses active field operations spanning the entire province of Misamis Oriental, whereas the Butuan branch scope is restricted to logistics and collection pipelines across the entire Caraga region. The system encompasses:

1. **Inventory Receiving:** Digital recording of incoming stocks based on delivery forms, with mandatory photo documentation and automated comparison against expected quantities.
2. **Inventory Releasing:** Recording of stocks released to specific agents, including QR code scanning for item identification, date stamps, product quantities, geotagged location data, and mandatory photo proof of handover.
3. **Stock Returns Management:** Tracking of unsold products returned by agents, with integration into inventory counts for future release.
4. **Stock Loss Recording:** Dedicated remarks section for documenting agent losses, as specifically requested by the client.

5. **Stock Monitoring:** Dashboard interface allowing managers to view current inventory levels remotely.
6. **QR Code-Based Stock Identification:** QR codes assigned to product batches to enable rapid scanning for receiving, releasing, and counting operations, reducing manual data entry errors and supporting FIFO compliance
7. **Geo-Tagging Functionality:** Automatic capture of GPS coordinates during stock release and receipt confirmation, providing a verifiable chain of custody for all inventory movements. This includes location stamping on transaction records and optional map display for management review, other than the aforementioned purposes, no continuous background tracking will occur, users are provided informed consent before enabling.
8. **Photo Handover Documentation:** Integration of camera functionality into release and receipt workflows to produce timestamped photographic evidence of every handover transaction, accessible only to authorized managers.
9. **Automated Weekly Reconciliation:** System-generated comparison of released stock against reported stock item sales and stock returns aligned to the company's weekly reporting cycle, with automatic flagging of discrepancies.
10. **User Role Management:** Different access levels for super admins, managers, sales representatives, collectors, and head office personnel.

ChemStock will be developed as a mobile application using React Native with the Expo managed workflow, targeting Android devices (version 10 / API 29 and above). This decision is based on the client's existing workforce device inventory, which consists exclusively of personal or company-provided Android devices, as no dedicated ICT hardware is currently available across branches. While an iOS version was initially considered for future scalability, development will be limited to Android only, as supporting iOS would incur additional costs without immediate necessity given the current user base. The system is not a web application or desktop software. The technical stack includes expo-sqlite for local database storage, Supabase for backend services, expo-camera for QR scanning and photo capture, expo-location for geolocation, NetInfo for offline synchronization, and expo-crypto for UUID generation.

This study acknowledges the following limitations:

1. **Geographical Scope:** The system will be developed and tested specifically for Cospachem's Butuan and Cagayan de Oro Branch (39 staff). While the design may be applicable to other branches among the 130+ nationwide, generalization would require additional adaptation and testing.
2. **Hardware Requirements:** The system assumes the users have access to smartphones. The company must provide the necessary device to the agents who lack personal smartphones. Geo-tagging functionality requires devices with enabled location services.
3. **Training Dependency:** Successful implementation depends on adequate user training. The study includes training recommendations but cannot guarantee uniform adoption or proficiency across all users.
4. **Privacy Considerations:** The collection of location data requires informed consent from all users. Geo-tagging will be limited to work-related transactions and will not involve continuous tracking of agent movements outside inventory-related activities.
5. **Payment Processing:** The system tracks receivables and collections but does not integrate with electronic payment gateways or banking systems. All financial transactions remain outside the system's scope.
6. **Complex Reporting:** While the system generates standard daily and monthly summary reports, highly customized ad-hoc reporting requests beyond the identified requirements may not be supported.

7. **No Physical Inventory Correction:** The system records and tracks inventory but cannot physically prevent loss or theft. It serves as an accountability tool rather than a physical security solution.
8. **Data Migration:** Historical inventory data exists only in paper format. Migrating this data to the new system would require manual entry, which is outside the scope of this development project. The system will focus on forward-looking transactions from implementation date onward.
9. **GPS Accuracy:** Geolocation data accuracy may vary depending on device quality, environmental factors (like tall buildings, indoor locations), and satellite availability. The system will record coordinates as captured without correcting or validating location accuracy.
10. **Testing Duration:** The evaluation period is limited to three weeks total (one week training + two weeks active testing). Long-term system effects on annual stock loss alarm and manager reimbursement cannot be fully validated within this timeframe.

1.6 Definition of Terms

- **Accountability.** The obligation of individuals to accept responsibility for inventory items released to them and to provide accurate accounting of those items' disposition.
- **Automated Reconciliation.** A system feature that automatically compares inventory releases against reported sales and returns to identify discrepancies.
- **Chain of Custody.** The step-by-step digital record of inventory movement, including geolocation data, timestamps, and user identification, establishing the sequence of custody from release to receipt.
- **Collector.** A field agent responsible for product batch delivery from branch to sales representative location.

- **Distributed Sales Model.** A business structure wherein sales representatives operate in assigned territories rather than from a central retail location.
- **First-In-First-Out (FIFO).** An inventory rotation method whereby the earliest acquired stocks are distributed first to minimize expiration risk.
- **Inventory Management System.** A technological solution for tracking inventory levels, orders, sales, and deliveries.
- **Geo-Tagging.** The process of embedding GPS-derived geographic coordinates into a digital transaction record at the moment the transaction occurs, creating a spatially verifiable audit trail for inventory movements.
- **MSME.** Micro, Small, and Medium Enterprise, as classified by the Philippine Statistics Authority based on asset size and employee count.
- **Inventory Visibility.** The ability to access current inventory information remotely with minimal latency following scheduled agent synchronization loops.
- **Photo Handover.** The systematic capture of timestamped photographic evidence during stock release and receipt activities, providing objective proof of what was transferred, in what condition, and by whom, to resolve accountability disputes.
- **QR Code (Quick Response Code).** A scannable qr code that stores product batch information readable by mobile device cameras. In ChemStock, QR codes are assigned to product batches to enable rapid scanning for receiving, releasing, and counting operations, replacing manual item-by-item data entry.
- **Sales Representative.** A field agent responsible for selling products directly to customers and remitting sales documentation to branch management.
- **Stock Loss.** Rarely tracked missing inventory, unreported sales, and occasional handover stock-count failures only become apparent during the annual branch audit.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the theoretical foundation and review of existing literature relevant to the development of ChemStock, a mobile-based inventory management system for Cospachem Products. It establishes the academic grounding for the study by examining theoretical frameworks, prior research on inventory management in Philippine MSMEs, existing technological solutions, and the specific gaps that ChemStock addresses.

2.1 Theoretical Framework

This study is anchored on the Task-Technology Fit (TTF) model developed by Goodhue and Thompson (1995), which provides a theoretical lens for understanding how technology leads to performance improvements. The TTF model explains that information systems positively impact individual performance when there is alignment between the capabilities of the technology and the requirements of the tasks that users must perform. The model identifies key constructs: task characteristics, technology characteristics, and task-technology fit, which together influence performance outcomes.

Task characteristics include the specific actions individuals perform to transform inputs into outputs. At Cospachem Products, inventory management involves a range of critical activities, including receiving stock deliveries and verifying quantities against orders, as well as releasing inventory to sales representatives, often without face-to-face interaction through channels like collectors. The process also requires tracking weekly sales and returns reported by agents, identifying and documenting losses at the individual agent level, and conducting periodic physical inventory counts. Finally, these tasks end with reconciling released stock against sales and returns to detect any discrepancies.

Technology characteristics refer to the tools, hardware, software, and data provided to assist users in performing their tasks. ChemStock's technology characteristics are anchored in a mobile-based platform designed to ensure accessibility for both office managers and sales representatives, thereby enabling critical inventory functions irrespective of geographical location. A major feature of this system is the integration of mobile GPS and geotagging functionalities, which capture precise location coordinates during both stock release and receipt confirmation. When inventory is sent via collectors without direct manager-agent interaction, the geotagging feature provides verifiable location data to confirm where and when the transaction took place.

To address the client's requirement for operational efficiency and accuracy during high-volume transaction periods, the system incorporates qr scanning functionality that automates inventory checking and mitigates manual data entry errors. Moreover, to resolve accountability gaps inherent in remote stock handovers, photo documentation is systematically integrated into release and receipt workflows, thereby providing objective proof of delivery and establishing a clear chain of custody. The system's core analytical capability resides in its automated reconciliation feature, which continuously compares stock releases against reported sales and returns to identify discrepancies upon data synchronization. Complementing this functionality is the dedicated agent-level loss tracking mechanism, which attributes inventory shrinkage to specific sales representatives through structured data fields, effectively transforming anonymous losses into actionable management information. Synthesizing these features, the system provides a dynamic dashboard that grants managers remote visibility into current inventory levels, facilitating proactive decision-making and enhanced operational oversight.

Task-technology fit is achieved when the capabilities of the technology directly address the specific requirements of the tasks it is designed to support. In the context of Cospachem Products, this alignment is demonstrated through several key features of the proposed system. First, the integration of photo documentation directly addresses the challenge of non-face-to-face stock releases by providing objective, timestamped proof of what items were dispatched, when, and to which sales representative, thereby resolving disputes and establishing accountability. Second, agent-level loss tracking addresses the critical task of identifying and attributing inventory shrinkage to specific individuals, enabling management to implement targeted accountability measures rather than treating losses as an unavoidable aggregate. Third, automated reconciliation, aligned with the company's weekly reporting cycle, addresses the task of prompt discrepancy detection, allowing the organization to identify and investigate variances in real time rather than waiting for annual head office audits. Finally, the real-time dashboard addresses the task of remote inventory management by eliminating the manager's reliance on manually captured document photographs while in the field, providing instant and accurate visibility into current stock levels across all branches and agents.

The TTF model also suggests that users will adopt and utilize a system when they perceive a good fit between task demands and system capabilities. This has implications for ChemStock's design: features must not only exist but be intuitively integrated into workflows so that users recognize their usefulness. The model thus serves both as a design framework (ensuring features match tasks) and as an evaluation framework (assessing whether users perceive the system as well-fitted to their work).

2.1.1 Organizational Structure and Role Definitions

Based on verification with the Branch Manager of the Cagayan de Oro branch, the organizational structure and role responsibilities for ChemStock implementation are defined as follows. Managers are responsible for processing the release of stocks to sales representatives and collectors, tracking inventory levels across all product categories, recording all inventory movements (including receiving, releasing, returns, and losses), checking release documentation to identify possible losses, and conducting inventory reconciliation and discrepancy investigations. Managers can create accounts for Sales Representatives and Collectors within their assigned branch using the system activation code provided by the Super Admin. Managers cannot create accounts for other Managers, this function is reserved exclusively for the Super Admin. Super Admin is a system-level user with exclusive rights to generate unique activation codes for Branch Manager account creation. These codes cryptographically authenticate and link each manager to their specific branch. The Super Admin does not create accounts for Sales Reps or Collectors, that is the Branch Manager's task after registration. The Super Admin has read-only access to system-wide summary reports for audit purposes and performs no inventory operations (e.g., releasing, receiving, QR generation, or transaction recording). This separation of duties prevents any single user from both provisioning manager accounts and manipulating inventory, enforcing checks and balances to deter fraud and ensure accountability. Sales Representatives are responsible for selling stocks in assigned geographical areas, requesting desired stock quantities from the manager, documenting sales transactions, returning unsold or leftover stocks to the branch, and confirming receipt of stock releases via the mobile system. Collectors function as stock couriers, delivering stock releases from the manager to the sales representative when face-to-face handover is not possible or when sales representatives cannot visit the branch office. Stock distribution follows two primary channels: face-to-face transactions at the branch office, or delivery via collectors. The verified branch office locations are as follows: Cagayan de Oro Branch at Blk 13 Lot 25, Malvar Street, Regency II, Iponan, Cagayan de Oro City; and Butuan Branch at Purok 2, Lumbocan, Butuan City.

2.1.2 Business Process Workflow (Current - Manual)

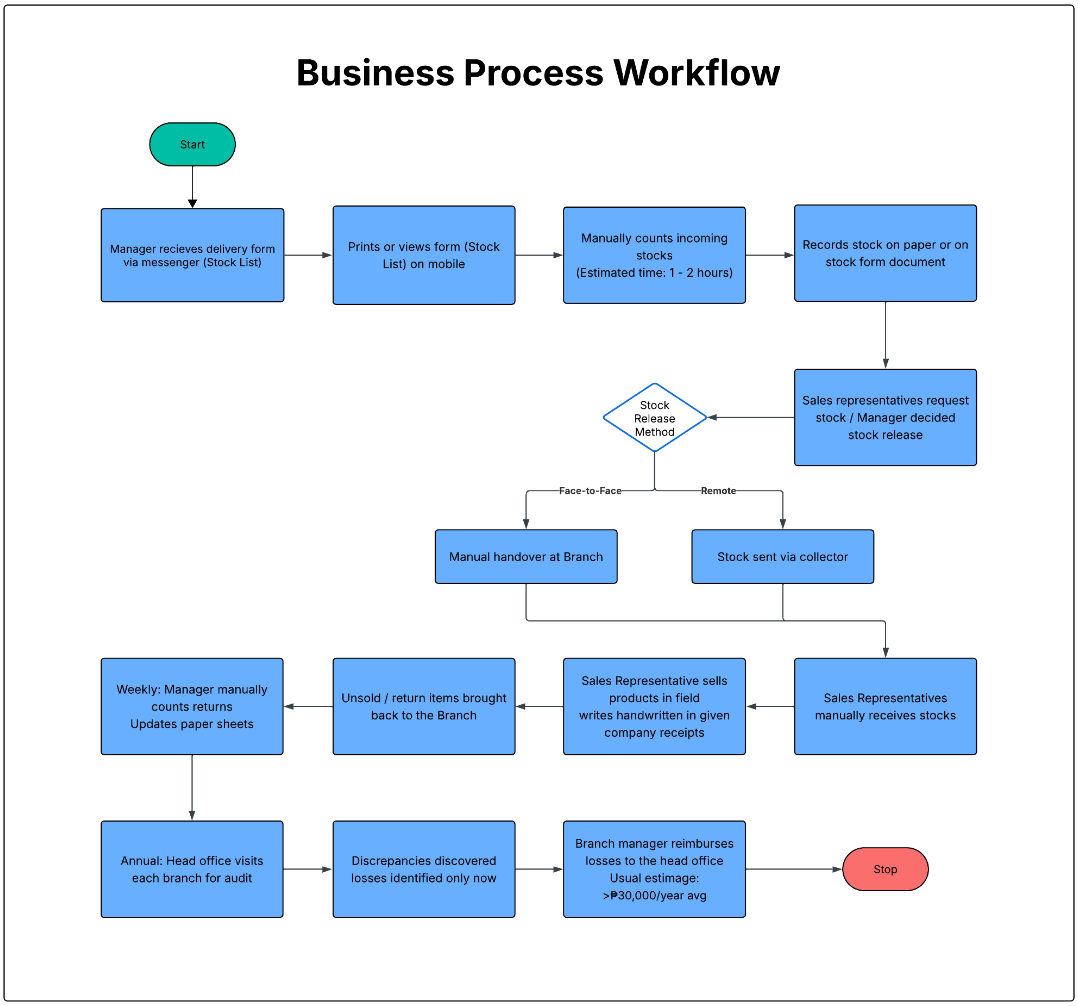


Figure 1. Business Process Workflow

Figure 1 presents the current business process workflow for inventory management at Cospachem Products, as documented through structured interviews with branch managers and staff. The process begins when the branch manager receives a delivery form via Messenger, which is either printed or viewed on a mobile device. Incoming stocks are then physically counted, a process that typically requires one to two hours per delivery. Stock counts are recorded manually on paper or stock form documents. When sales representatives request stocks or when the manager decides on a stock release, the transaction occurs either through face-to-face handover at the branch or remotely via collector. Sales representatives receive the stocks manually and proceed to sell products in the field, issuing handwritten company receipts. Unsold or returned items are brought back to the branch, where the manager manually counts returns and updates paper sheets on a weekly basis. The cycle culminates in an annual head office visit to each branch for audit, at which point inventory discrepancies and stock losses are identified. Subsequently, the branch manager is required to reimburse the head office for verified losses, with annual reimbursements typically exceeding 30,000 Philippine Pesos. This workflow reveals multiple points of inefficiency and accountability gaps, which the proposed ChemStock system aims to address through digitization, real-time tracking, and verifiable chain of custody mechanisms.

2.1.3 Proposed System Workflow (ChemStock - Automated)

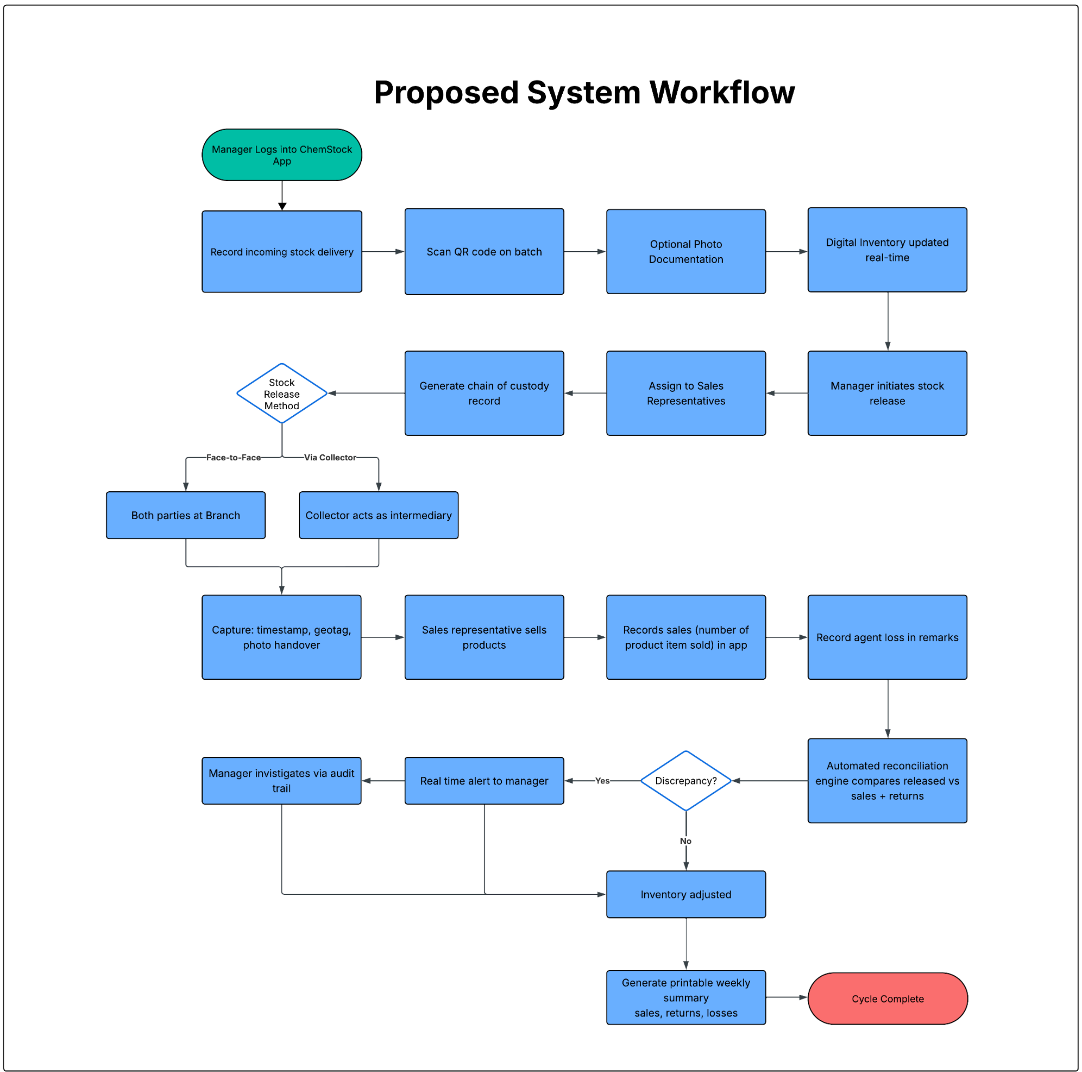


Figure 2. Proposed System Workflow

Figure 2 presents the proposed business process workflow for the ChemStock automated inventory management system. The process begins when the manager logs into the ChemStock mobile application and records an incoming stock delivery by scanning the QR code on each product batch, with a mandatory photo documentation step. The digital inventory is then updated upon synchronization. To release stock, the manager initiates the release process, assigns the stock to specific sales representatives, and generates a verifiable chain of custody record. Depending on the release method, either face-to-face at the branch or via a collector acting as an intermediary, the system captures a timestamp, geotagged location, and photo handover documentation at the point of transfer. The sales representative proceeds to sell products in the field and records the number of items sold directly in the application, while any agent losses are documented in a dedicated remarks field. An automated reconciliation engine continuously compares released stock against reported sales and returns. When a discrepancy is detected, an automated alert is sent to the manager, who can investigate the issue through the system's audit trail. When no discrepancy exists, the inventory is automatically adjusted. The cycle concludes with the generation of a printable weekly summary of sales, returns, and losses, providing management with complete and timely visibility into branch operations.

2.2 Traditional Inventory Management in Philippine MSMEs

Inventory management in Philippine micro, small, and medium enterprises is largely informal and deeply embedded in day-to-day operational realities. Despite the availability of digital solutions and clear legal frameworks governing business operations, the sector often operates through manual, paper-based systems that introduce vulnerabilities including human error, data loss, and inefficient resource allocation.

2.2.1 Manual Recording Practices and Their Limitations

In many Filipino MSMEs, pen-and-paper remains the primary method for inventory tracking. Brela, Ochua, Gabriel, and Resuello (2024) conducted research on micro-small construction supply stores and found that manually recording inventory and sales-related data is labor-intensive, paper invoices are at risk of misplacement and susceptible to fraudulent transactions, and lack of transparency in inventory levels can result in overstocking or understocking issues. Their study, published in the ACM Digital Library, employed a mixed-methods approach combining qualitative and quantitative data to identify distinct user requirements including real-time inventory tracking and intuitive user interfaces accessible to all employees.

This pattern is closely tied to resource constraints common among Philippine MSMEs. Ines (2022) explored financial management and performance of small and medium enterprises in Cauayan City, concluding that SMEs most often practice recording and budgeting, while inventory management is practiced only sometimes, primarily due to the small scale of operations and limited resources. The study found that inventory management is significantly related to financial statements, meaning that the larger the income earned by the business, the more significant the variation in the manner of managing physical stocks.

2.2.2 Distributed Sales Models and Accountability Challenges

For businesses operating with distributed sales forces, where inventory is released to field agents rather than held in a central retail location, accountability challenges multiply. Mostiero (2022) examined working capital management practices of microenterprises in Lipa City as predictors of business sustainability. The study documented that businesses often minimize inventory shrinkage by preparing ending inventory reports and applying different inventory management techniques such as having CCTVs in place and observing the FIFO method. However, these measures are less effective when inventory physically leaves the premises with agents who operate independently in the field.

The releasing and distribution of stock to field agents creates particular accountability concerns. Mostiero (2022) notes that inventory shrinkage, the loss of products between point of manufacture and point of sale, represents a significant challenge for businesses with distributed operations. Without systematic documentation of who received what, when, and in what condition, disputes become inevitable and losses become invisible until inventory counts reveal discrepancies.

2.2.3 The Digital Divide in Philippine MSMEs

Macatumbas-Corpuz and Bool (2021) studied Barangay Micro Business Enterprises (BMBEs) in Ilocos Norte and found that working capital and inventory management practices were significantly linked to business sustainability outcomes. Their research demonstrated that while MSME owners generally recognize the importance of structured inventory practices, actual implementation of systematic methods remains inconsistent, particularly among micro-scale operators. However, despite this awareness, adoption of digital solutions remains limited, often attributed to resource constraints and the perceived complexity of available tools.

Hines (2020) explored the utilization of technology in small businesses, noting that even though the number of small businesses far surpasses larger firms and their contribution to the economy is significant, very few studies have examined small businesses' use and adoption of technology. The adoption of technology by small businesses could be one key factor that could keep these organizations profitable and competitive in the changing world of business.

This digital divide manifests across three distinct dimensions identified by Hines (2020) in a study on small business technology adoption: awareness, where business owners lack knowledge of available digital solutions; willingness, where perceived cost or complexity discourages adoption; and capability, where limited resources, infrastructure, or technical support prevent implementation. Hines (2020) further emphasizes that despite small businesses' significant economic contributions, technology adoption remains disproportionately low relative to their numbers. For Cospachem Products, this framework explains why manual systems persist despite their documented limitations, and underscores the importance of designing ChemStock for ease of adoption by users with limited technical backgrounds.

2.3 Technologies and Existing Systems

The rise of digital platforms has transformed how businesses manage inventory and operations. In the Philippines, both commercial inventory systems and custom-developed solutions are used to track stock, manage releases, and monitor accountability. This section reviews existing systems and research relevant to ChemStock.

2.3.1 Commercial Inventory Management Systems

Odoo Inventory (Odoo S.A., 2023) offers a comprehensive warehouse management system designed to maximize efficiency. Features include real-time inventory tracking, automated replenishment, qr scanning, and integration with other business applications. The system is designed for businesses of various sizes and provides a modular approach that allows companies to implement only the features they need. However, Odoo is designed primarily for centralized warehouse operations rather than the specific needs of a distributed sales force where inventory is entrusted to remote agents.

SkuNexus (2023) provides a cloud-based warehouse management system offering real-time inventory visibility, scalability, flexibility, and cost efficiency. The platform features real-time operations management, automated workflows, and integration capabilities with existing ERP systems, accounting software, and e-commerce platforms. Key benefits include streamlined operations and automation, real-time inventory visibility, and scalability as businesses grow. Case studies documented by SkuNexus include Graeter's Ice Cream achieving 100% order automation with real-time inventory management, demonstrating the practical benefits of cloud-based systems.

Zoho Inventory (Zoho Corporation, 2023) offers online inventory management software designed for small and medium businesses. Features include order management, inventory tracking, warehouse management, and shipping integrations. The system provides real-time stock updates across multiple sales channels and supports qr scanning for efficient inventory operations.

Sortly (Sortly, 2023) is a visual inventory management system that emphasizes photo documentation and ease of use. It allows users to attach photos to items, track quantities, and manage inventory across locations. While Sortly supports photo documentation, it is designed for asset tracking rather than the workflow of releasing inventory to field agents with delayed reconciliation.

While these commercial examples demonstrate the practical applications and benefits of cloud-based inventory systems, they are designed primarily for warehouse, retail, or asset management rather than the specific needs of a distributed sales force. They lack workflow-integrated features for remote handover proof, agent-level loss attribution, and reconciliation aligned with weekly reporting cycles. These capabilities are required to manage operations where inventory is physically separated from management oversight.

2.3.2 Philippine Academic Research on Inventory Systems

Several academic studies from Philippine institutions have explored the development and implementation of inventory management systems tailored to local business contexts.

Moreno, Javier, Francia, and Harina (2023) investigated the mediating effect of replenishment decisions on cloud-based inventory management and record keeping performance among microbusinesses in Laguna, Philippines. Published in the ACM International Conference Proceedings, this peer-reviewed study had a sample size of 52 respondents and found that utilizing cloud-based systems results in more precise inventory records and better ability to maintain necessary supply levels. The research employed the Task-Technology Fit (TTF) framework and concluded that cloud-based inventory management significantly improves record keeping performance when replenishment decisions are properly managed.

Intal, Miranda, Sitoy, and Matoza (2022) developed iBiz, an Android mobile application with business intelligence for retail micro and small enterprises. Published in the ACM International Conference Proceedings, the study used Dart and Flutter SDK with Firebase and Cloudinary back-ends. The application was tested with 40 MSE owners through usability surveys, with Likert scale analysis showing "Excellent" ratings for functionality. Respondents "Strongly Agree" to importance and willingness to use the application, demonstrating successful mobile application development for Philippine MSEs with strong user acceptance.

Cruz and Villanueva (2023) studied the design and implementation of an inventory monitoring system with qr technology specifically for micro enterprises. Their research demonstrated that qr integration significantly reduced data entry errors and accelerated inventory counting processes. The study found that microenterprise operators could be trained effectively on qr-based systems and that such technology was appropriate even for businesses with limited technical resources. This directly supports the client's affirmative response to whether scanning boxes using a mobile phone would help speed up the checking process and validates the inclusion of qr scanning functionality in ChemStock.

Santos and Reyes (2023) explored production monitoring and inventory control systems for small-scale manufacturing through a case study in the Philippines. Their research demonstrated that integrated inventory systems significantly reduce manual errors and improve operational efficiency. The study found that small-scale enterprises implementing digital inventory solutions experienced a marked decrease in stock discrepancies and improved production planning capabilities.

2.3.3 International Studies on Inventory Systems

Medina-Tapara, Orihuela-Vera, and Cabanillas-Carbonell (2024) developed and evaluated a mobile system with usage pattern analysis to optimize storage management. Published in the International Journal on Advanced Science, Engineering and Information Technology, this study demonstrated significant benefits including 9.85% efficiency improvement from mobile app implementation and 10.18% cost reduction. The study was evaluated by 10 experts using Likert scale on usability, design, functionality, and efficiency, with positive impact on product distribution.

Panigrahi, Shrivastava, and Kapur (2023) conducted a comprehensive systematic literature review examining the connection between inventory management practices and operational performance of SMEs from 2000-2023. Published in the International Journal of System Assurance Engineering and Management, this review identified five clusters, seven thematic areas, and four emerging research contents. The study integrated bibliometric analysis to show how IoT, big data, and machine learning technologies impact SME inventory cost reduction.

2.4 QR Technology, Real-Time Visibility, and Accountability Features

Modern inventory management systems increasingly rely on qr technology and real-time visibility to enable proactive decision-making and error reduction. Additionally, features such as photo documentation address accountability gaps in distributed operations.

2.4.1 QR Scanning for Error Reduction

Qr technology transforms inventory counting from a manual, error-prone process into an efficient, accurate operation. Cruz and Villanueva (2023) demonstrated that qr integration significantly reduced data entry errors and accelerated inventory counting processes for micro enterprises. Their research found that operators could be trained effectively on qr-based systems and that such technology was appropriate even for businesses with limited technical resources.

The client's response to whether scanning boxes using a mobile phone would help speed up the checking process ("Yes, that would be fine and more helpful") validates the inclusion of qr scanning functionality in ChemStock. Similarly, the acknowledgment that qr scanning would reduce errors compared to manual input supports the technological direction of the proposed system.

2.4.2 Real-Time Inventory Visibility

Research has identified that among the distinct user requirements for inventory systems are the need for real-time inventory tracking, an intuitive user interface accessible to all employees, and efficient tracking of items with varying characteristics (Brela et al., 2024). Cloud-based warehouse management systems provide real-time operations management, offering instant updates from inventory changes to order status, accessible from anywhere with an internet connection (SkuNexus, 2023).

For Cospachem, where managers are not always face-to-face with sales representatives when releasing stocks and sometimes send items via collectors, real-time visibility would provide crucial oversight. The client's expressed need for real-time stock monitoring while in the field is currently addressed by an inefficient workaround: she "takes pictures of the sales and stock records to review later," accumulating numerous photos that are only deleted after verification.

Mendoza and Fernandez (2023) developed and evaluated a web-based logistics and inventory management system specifically designed for small and medium enterprises. Their research demonstrated that real-time inventory visibility combined with logistics tracking significantly improved operational efficiency for SMEs. The study found that MSMEs implementing such systems experienced enhanced decision-making capabilities, reduced stockouts, and improved customer satisfaction.

2.4.3 Photo Documentation and Geotagging for Accountability

The client affirmed that requiring quick photos during receiving or releasing would help provide proof, "especially since doing it on a phone is handy." Recording date, time, and employee handling was described as "YES, would be very helpful." These requirements directly inform the accountability features needed in the proposed system.

Photo documentation addresses the specific challenge of non-face-to-face stock releases. When items are sent via collectors, photographic evidence of the package, its contents, and its condition at the time of release provides objective proof that can resolve disputes about what was delivered. This feature, combined with digital timestamps and user identification, establishes a clear chain of custody for every inventory movement.

Beyond visual documentation, geotagging, the embedding of GPS coordinates into transaction records, adds a critical spatial dimension to accountability. When a manager releases stocks and the system automatically captures the release location, and when an agent acknowledges receipt with a simultaneous capture of their location, the system creates a geographic audit trail that prevents agents from falsely claiming they were elsewhere during a scheduled delivery. This dual-location capture addresses a specific vulnerability in Cospachem's operations, specifically the absence of face-to-face interaction during stock handovers. A dispute wherein an agent claims to have never received stocks becomes objectively resolvable when the system shows that the agent's device recorded the receipt confirmation at a location matching the agent's assigned territory at a time consistent with the delivery schedule.

Research supports the effectiveness of location-based accountability features in inventory management contexts. Medina-Tapara, Orihuela-Vera, and Cabanillas-Carbonell (2024) demonstrated that mobile systems incorporating usage pattern analysis and location tracking improved operational efficiency by 9.85% and reduced costs by 10.18% in distribution operations. The study's findings indicate that location data, when captured at transaction points rather than continuously, provides sufficient accountability without compromising user privacy. Similarly, Cruz and Villanueva (2023) noted that micro-enterprises employing mobile-based tracking systems experienced fewer disputes regarding inventory transfers compared to those relying solely on paper documentation.

Nachiappan, Jawahar, and Prasanna (2021) further examined GPS-stamped delivery confirmation in last-mile distribution operations and found that scan-event logging with location capture substantially reduced accountability disputes between dispatchers and delivery agents, because the system produced an objective record that neither party could retroactively modify. This finding directly parallels the accountability challenge at Cospachem, where Sales Representatives occasionally deny receiving products and no objective spatial record currently exists to resolve such disputes.

Carrier, Hasan, and Youn (2022) established that in distributed operations where goods change hands across geographic distances, the combination of photographic evidence and GPS location data constitutes the minimum evidentiary standard for a defensible custody record. This is particularly relevant for Cospachem's collector-based delivery channel, where goods transit through third-party carriers without any currently documented chain of custody. ChemStock's implementation of photo plus GPS capture at both the release and receipt endpoints transforms this channel from an accountability gap into a two-point auditable record.

For Cospachem, the combination of photo documentation and geotagging serves a dual purpose: visual proof establishes what was delivered, while location proof establishes where delivery and receipt occurred. Together, these features transform what was previously an unverifiable process into auditable digital evidence, directly addressing the client's stated need to maintain a "standardized format that includes the list of items, release date, sales, returns, and a remarks section to indicate any losses" with geotagging adding the critical dimension of where these transactions took place.

2.5 QR Code Technology in Inventory Management

Quick Response (QR) codes are two-dimensional matrix qr readable by standard smartphone cameras without specialized hardware, making them particularly suited for mobile-first inventory systems in resource-constrained environments. While Cruz and Villanueva (2023) established that qr technology generally reduces errors in micro-enterprise inventory management, QR codes extend this capability by encoding richer product information, including batch identifiers and sequential stock numbers, within a single scannable code. Aziz, Othman, Razak, and Harun (2019) demonstrated that QR code-based inventory systems significantly reduced data entry time and minimized manual recording errors compared to paper-based methods, with transaction processing time reduced by approximately 40 percent. For Cospachem, where each release involves multiple product types that must be individually counted, a QR code assigned to a product batch allows a Sales Representative to confirm an entire release through a single scan rather than manually recording each item, directly addressing Problem D of this study.

Beyond error reduction, QR scanning creates a digital audit trail that timestamps exactly when an item was scanned and by which user account, and when combined with GPS capture, at what location. Nachiappan, Jawahar, and Prasanna (2021) found that scan-event logging in last-mile distribution operations substantially reduced disputes between dispatchers and delivery agents, because the system produced an objective timestamped record that neither party could retroactively modify. This directly parallels the accountability challenge at Cospachem. The suitability of QR technology for Philippine MSMEs is further confirmed by Intal, Miranda, Sitoy, and Matoza (2022), who found that mobile scanning features received the highest usability ratings among all features evaluated in their iBiz application study, and by the Cospachem office manager's direct confirmation that QR scanning would be "fine and more helpful" and would "reduce errors compared to manual input."

2.6 Chain of Custody in Inventory and Logistics Systems

Chain of custody refers to the documented chronological record of who possessed, transferred, or handled an item at every point in its movement from origin to destination. In inventory and logistics management, the concept describes the complete, verifiable sequence of custodial transfers for a physical good from the point of storage through to its final sale or return. Carrier, Hasan, and Youn (2022) define supply chain custody documentation as the systematic capture of transfer events, including the identities of transferring and receiving parties, timestamps, locations, and condition evidence, sufficient to reconstruct the full movement history of any item and assign responsibility at each stage.

Despite this established framework, existing commercial inventory systems such as Odoo, Zoho Inventory, Sortly, and SkuNexus do not implement chain of custody mechanisms for distributed sales operations. These systems are designed for centralized warehouse environments where inventory remains under continuous management oversight, and they provide no mechanism to verify that a transfer occurred between a specific releasing party and a specific receiving party at a specific location. This is a critical gap when inventory is dispatched via collectors without face-to-face interaction, as is the case at Cospachem Products. Mostiero (2022) identified precisely this documentation failure as a primary driver of inventory shrinkage in distributed operations, noting that the absence of systematic transfer records makes it structurally impossible to determine whether losses occur in transit, upon non-receipt by the agent, or after receipt. ChemStock directly addresses this gap by capturing a timestamped, GPS-stamped, and photo-evidenced custody record at both the release and receipt endpoints of every inventory transfer, transforming an inherently unverifiable handover process into a two-point auditable chain of custody.

2.7 Research Gap

Existing literature establishes three key findings relevant to this study. First, manual inventory systems in Philippine MSMEs are prone to errors, losses, and accountability gaps. Studies by Brela et al. (2024), Ines (2022), and Mostiero (2022) document the limitations of paper-based methods, including labor intensity, risk of fraud, lack of transparency, and undetected shrinkage. Second, cloud-based inventory systems improve record accuracy and operational efficiency when properly implemented. Moreno et al. (2023) demonstrated that cloud-based systems enhance record keeping performance, while Intal et al. (2022) and Cruz and Villanueva (2023) showed that mobile and qr-based solutions are acceptable and effective in Philippine micro-enterprises. Third, commercial systems like Odoo, Zoho, and Sortly offer robust inventory features but are designed for warehouse, retail, or asset management contexts. These systems support qr scanning, real-time visibility, and even photo documentation (in Sortly's case), yet they are not optimized for the specific workflow of a distributed sales force where inventory is released to remote agents with delayed reconciliation.

While these findings provide a valuable foundation, they collectively overlook the unique constraints of distributed sales operations. Consequently, four specific gaps remain unaddressed in the literature. First, the adaptation of digital inventory systems for distributed sales models remains unexplored. Existing studies focus on either manual systems or commercial solutions but do not examine how digital systems must be adapted for operational structures where inventory physically separates from management, leaving the unique challenges of non-face-to-face releases, agent-level loss attribution, and weekly reporting cycles unaddressed. Second, there is a lack of proposed and evaluated technological mechanisms for agent accountability. While Mostiero (2022) identifies accountability as a challenge in distributed operations, no studies propose or evaluate specific technological mechanisms, such as photo documentation embedded in release workflows or agent-level loss tracking, designed to address this challenge. This results in a literature that lacks design principles for accountability features tailored to field-based sales forces. Third, reconciliation features aligned with periodic reporting cycles are absent. Existing research and commercial systems assume real-time reconciliation, yet many MSMEs operate on weekly or monthly remittance cycles where delayed reconciliation is an operational reality, not a design flaw. Fourth, no existing system integrates QR-based stock identification with geo-tagged chain of custody for distributed handovers. This combination is essential when inventory is transferred through third-party intermediaries such as collectors without face-to-face verification. Aziz et al. (2019) and Nachiappan et al. (2021) demonstrate QR tracking's value in logistics, and Carrier et al. (2022) establish the evidentiary standard for custody records, yet no study or commercial system combines these capabilities into a workflow designed for MSME distributed sales operations. Together, these gaps highlight the absence of inventory solutions tailored to the distributed sales model.

To address these gaps, this study designs ChemStock, a system whose features are deliberately configured for the distributed sales model. Photo documentation is embedded in release workflows to provide proof for remote handovers. Agent-level loss tracking with dedicated fields enables attribution of shrinkage to specific individuals. QR code-based stock identification replaces manual item counting with a single-scan confirmation per batch. Automated reconciliation is aligned with weekly reporting cycles, comparing releases against reported sales and returns with appropriate time delays. In doing so, this study contributes to the literature on MSME digital transformation by demonstrating how generic inventory capabilities must be reconfigured for specific operational contexts, and by proposing design features that enhance accountability in distributed sales operations.

2.8 Comparison of Existing Systems

Functionality	**ChemStock (Proposed)**	**Zoho Inventory**	**Sortly**	**Odoo Inventory**	**SkuNexus**
Real-Time Inventory Visibility	✓	✓	✓	✓	✓
Photo-Proof of Internal Handover	✓	X	X	X	X
Dedicated Agent Loss Recording	✓	X	X	X	X
Automated Weekly Reconciliation	✓	X	X	X	X
Distributed Agent Release Workflow	✓	X	X	X	X
Geotagged Transaction Proof	✓	X	X	X	X
QR-Based Stock Identification	✓	X	X	X	X

Table 2. Comparison of Existing Systems

CHAPTER III

METHODOLOGY

This chapter describes and discusses the development methodology and process used in building the ChemStock system. In addition, it also explains how the researchers gathered the necessary data and information used in the design and implementation of the system. It includes the respondents involved in the requirement gathering and system evaluation. This also shows the procedure of data collection and the instruments used. In addition, this chapter discusses the system design, the technical stack selected, and the research locale where the study was conducted.

3.1 Project Development Methodology

Figure 3 shows the Agile Development model, which employs a flexible and iterative methodology that facilitates ongoing adaptation to evolving project requirements and incorporates user feedback throughout the development lifecycle. This model is specifically chosen due to its emphasis on continuous collaborative work with the client, developers and stakeholders, ensuring effective communication and gathering of user requirements. Agile methods are widely recognized as a natural fit for mobile application development because their iterative cycles accommodate changing requirements, reduce time to market, and ensure that the final software aligns perfectly with immediate user needs (Kaleel & Harishankar, 2013). Through cross-functional collaboration and iterative development practices, the research team performed continuous assessments and addressed emerging issues in a timely manner, allowing the system to adapt incrementally to user requirements. This approach facilitates the delivery of operational software within a two-week timeframe while preserving high levels of usability and functional integrity. The inherent flexibility of Agile methods balanced adaptability with operational efficiency, ensuring that the final application remained aligned with user expectations.

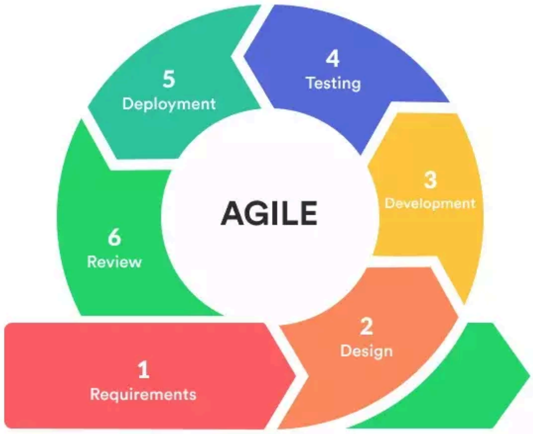


Figure 3. Agile Development Cycle

3.2 Requirement Gathering and Analysis

This chapter discusses the procedure and method used to collect and analyze the Chemstock requirements. It contains information about the research local, respondent of the study, and the research instrument used.

3.2.1 Research Local

The study was conducted in Cagayan De Oro City, Philippines. Respondents were interviewed both in person at their office situated at Blk 13 Lot 25, Malvar St. Regency II A, Iponan, Cagayan De Oro City, and through scheduled online meetings via Google Meet. To properly evaluate the mobile system's interval-based tracking capabilities under real-world field conditions, the research locales are defined by the complete administrative jurisdictions of both hubs. The field territory for the Cagayan de Oro locale spans the expansive geographic boundaries of Misamis Oriental, while the field territory for the Butuan locale covers the entirety of the Caraga region. These vast regional testbeds provide the ideal environment to validate the system's spatial chain-of-custody tracking.

To ensure a highly controlled evaluation environment across these vast regional testbeds, both the Cagayan de Oro and Butuan branches operate under a unified management structure led by the same regional branch manager. This dual-branch administrative configuration provides a unique advantage for the study's internal validity, as it guarantees that operational workflows, accountability standards, and business process compliance remain entirely identical across both testing environments. Furthermore, this shared leadership perfectly highlights the critical need for an on-demand digital inventory platform, as a single manager must oversee sprawling, independent field operations across both Misamis Oriental and the Caraga region simultaneously without the Background of the Study possibility of constant physical presence at either branch office.

3.2.2 Respondent of the Study

The respondents for this study compromise the primary user groups interacting with the ChemStock mobile application. These are categorized as follows:

Super Admin: This category comprises a system-level user who does not engage with daily inventory operations but instead oversees user account governance and branch-wide performance monitoring. The Super Admin’s responsibilities include managing all system users account across branches (creating, editing, deleting, and assigning or revoking user roles) and viewing system-wide report summaries and alerts dashboards for oversight purposes.

Manager: This category comprises users who engage with ChemStock's inventory tracking capabilities, QR-based batch release functions, stock release mechanisms, and return acceptance features. They include:

- **Branch Managers:** Managers of a specific branch who utilize the platform to track stock inventory and agent loss, execute QR-based batch release functions, manage stock release mechanism, and process return acceptance features.

Sales Representative: This category comprises users who engage with ChemStock's agent inventory tracking capabilities, and stock request functions, stock return features. They include:

- **Regular Sales Representatives:** Sales personnel permanently assigned to specific physical areas who manage routine inventory tracking, stock request fulfillment, and return processing for their designated agent accounts.
- **Short-Term Sales Representatives:** Sales personnel temporarily assigned to cover specific physical areas on an ad-hoc basis still having the same platform features as Regular Sales Representatives.

Collectors: This category comprises users who engage with ChemStock's stock delivery function. They include:

- **Regular Collectors:** Collection agents permanently assigned to collect outstanding stock balances and facilitate bulk stock deliveries from the branch office to sales representatives when in-person stock release is unavailable.
- **Short-Term Collectors:** Collection agents temporarily assigned to perform the same duties, collecting outstanding stock balances and handling bulk stock deliveries from the branch

office to sales representatives, when no regular collector is available.

3.2.3 Data Collection and Analysis Procedure

Interviews (via Google Meet) were recorded with consent, transcribed, and analyzed using thematic analysis. Requirements were extracted, categorized by feature, and validated with two branch managers before finalization.

3.3 System Design

3.3.1 System Architecture

Figure 4 illustrates the system architecture of ChemStock. Branch Managers, Sales Representatives, and Collectors access the platform through the presentation layer, which is a React Native Expo mobile application targeting Android devices. This layer integrates native device hardware, including the camera for QR scanning and photo documentation, and GPS for geotagging, ensuring that every transaction is recorded with a verified location. Generated QR codes can also be printed or shared directly from the application. When internet connectivity is unavailable, the presentation layer stores transaction records locally via SQLite and queues them for synchronization once the connection is restored.

All client requests are routed through the internet to the application layer, which is a centralized Express JS RESTful API. This layer serves as the system's business logic hub, processing every inventory event from stock receiving and releasing to returns and loss recording while enforcing role-based access control for each user type. The application layer also hosts the automated reconciliation engine, which continuously compares released stocks against reported sales and returns, and triggers real-time discrepancy alerts to the Branch Manager when variances are detected.

The backend layer is composed of three storage components. PostgreSQL via Supabase serves as the primary cloud database, storing all centralized inventory records, user profiles, transaction logs, GPS coordinates, and alert data, while its real-time subscription capabilities enable instant dashboard updates across all users. Supabase Bucket Storage handles all photo and media evidence captured during handovers, linking each timestamped image to its corresponding transaction record. SQLite via expo-sqlite functions as the local on-device database, temporarily holding transaction records when the device is offline and automatically synchronizing them to Supabase once internet connectivity is restored.

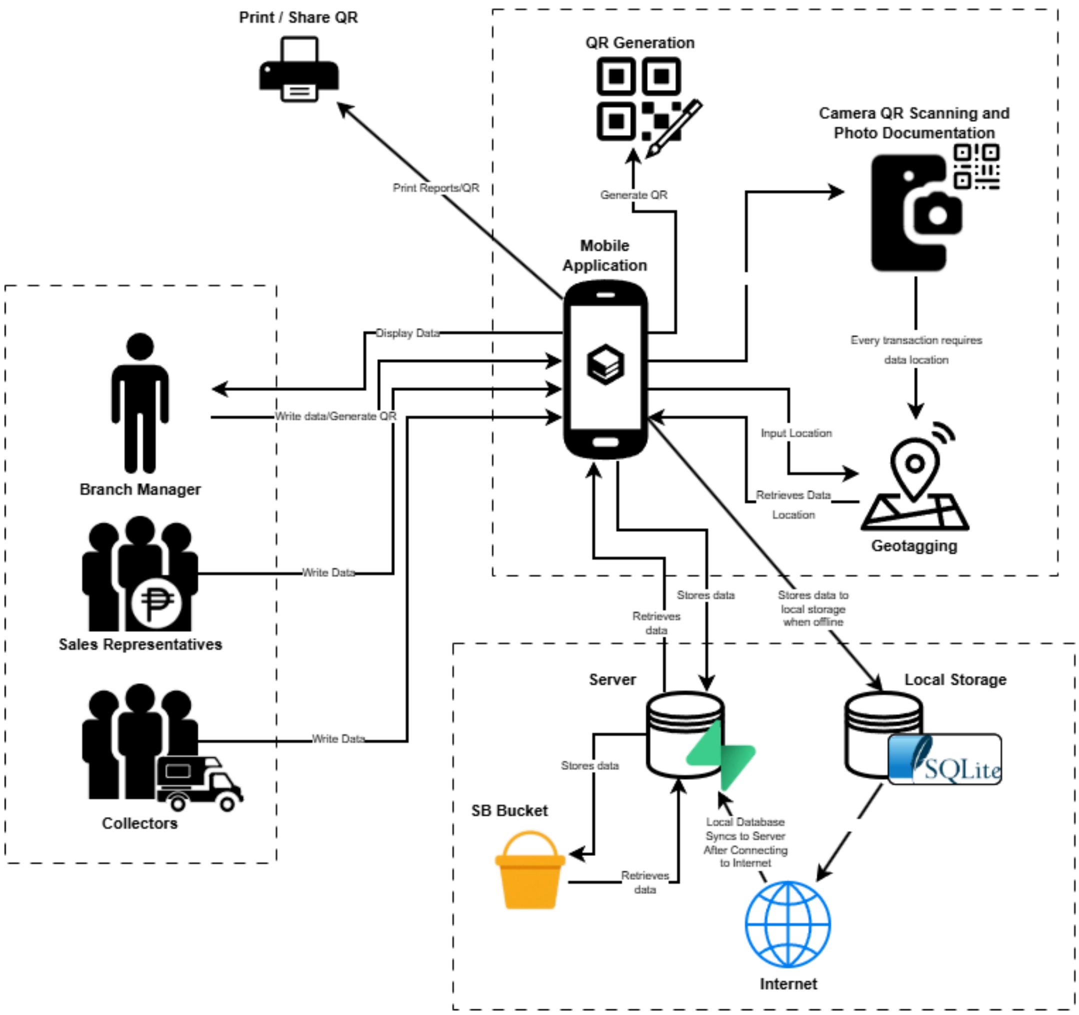


Figure 4. System Architecture for ChemStock

3.3.2 Use Case Diagram

Figure 5 shows the functions and interactions of the three main users within the ChemStock system: the Branch Manager, Sales Representative, and Collector. The Super Admin has two main functions: logging in and generating activation codes for Branch Manager account creation. These codes enable Branch Managers to register and then provision accounts for Sales Reps and Collectors in their branch. The Super Admin does not create user accounts directly or handle any inventory operations. This duty separation prevents any single user from both provisioning manager accounts and manipulating inventory, ensuring accountability and fraud prevention. The Branch Manager has the widest range of functions, which include logging into the system, creating accounts and roles, creating QR codes and recording stocks, releasing stocks, tracking inventory, and viewing reports and alert summaries. Sales Representatives can perform stock requests, confirm stock receivables, record transactions, and accept stock deliveries. Collectors share several of the same functions as Sales Representatives, including confirming stock receivables, recording transactions, and accepting stock deliveries. Two important dependencies are shown in the diagram. First, the Releasing of Stock function carries an <<include>> dependency on QR Scanning, making QR code verification a required step in every stock release, while Capture Photo Proof is connected through an <<include>> relationship that allows mandatory photographic documentation to be added as supporting evidence during the handover. Second, both the Confirm Stock Receivables and Accept Stock Deliveries functions carry an <<include>> dependency on Capture GPS Location, meaning that every field transaction performed by the Sales Representative and Collector must automatically record the GPS coordinates at the exact time and point of the activity, ensuring that all stock movements are supported by a verified location record.

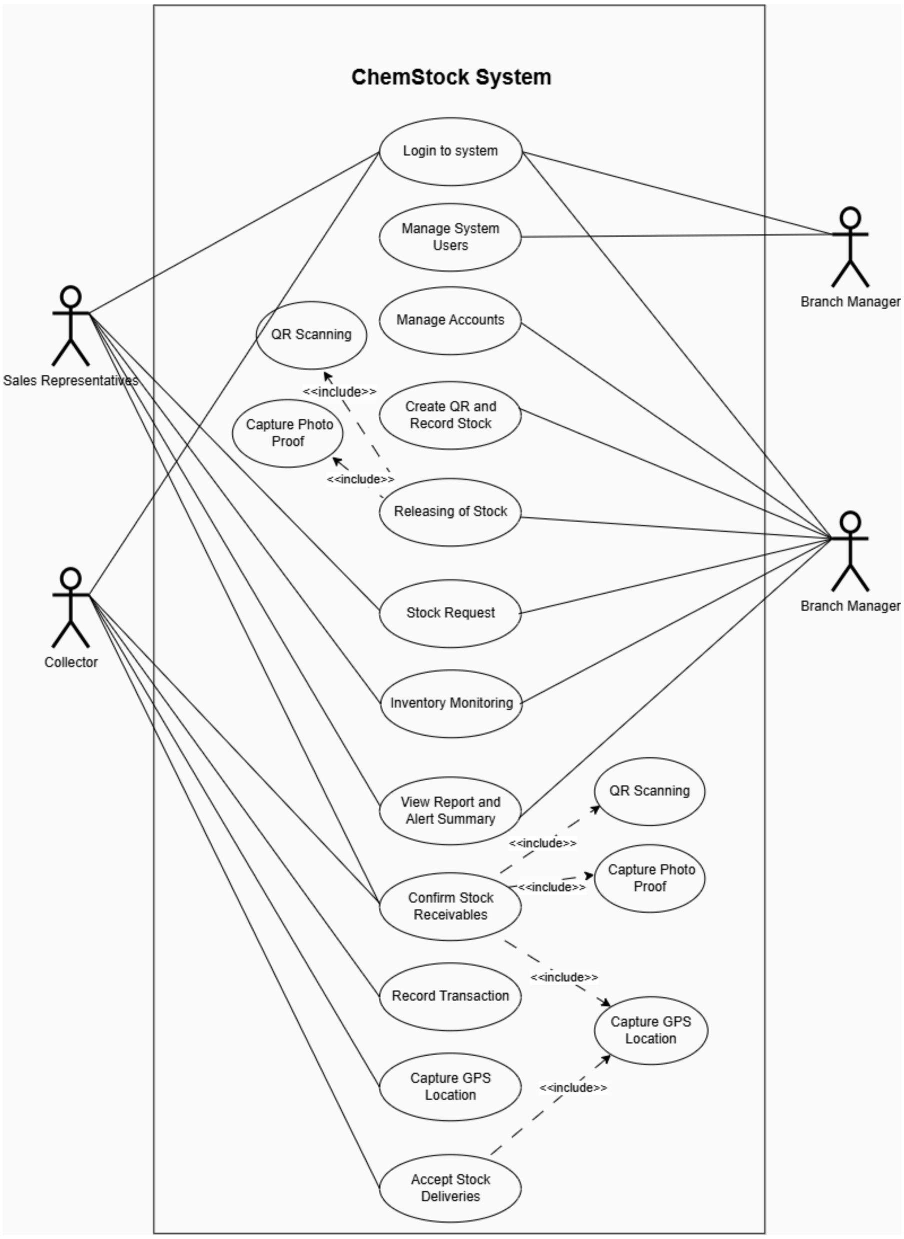


Figure 5. Use Case Diagram for ChemStock

3.3.3 Context Diagram

Figure 6 illustrates the context diagram of ChemStock, defining the system boundary and the data flows between the ChemStock system and its three external entities: the Manager, the Sales Representative, and the Collector.

The Super Admin is a system-level entity with limited but critical governance and access-control duties. Inbound: receives requests for activation codes used by Branch Managers to register. Outbound: provides unique cryptographic codes for that purpose. The Super Admin does not create accounts for Sales Reps or Collectors, that is delegated to Branch Managers, nor does it perform any inventory operations. This ensures a strict separation between system governance and inventory management.

The Manager, as the primary administrative entity, with inbound data flows to the ChemStock system including stock release governance instructions, QR code production requests, user registration details, incoming stock records, return management dashboard entries, and agent loss records and remarks; and outbound data flows from the ChemStock system to the Manager including stock request confirmation details, return request confirmation details, remote inventory monitoring information, reports and alert summaries, and a delivery summary for management decision-making.

The Sales Representative, with inbound data flows to the ChemStock system consisting of stock request information, delivery receipt confirmation details with photoproof and GPS location, return request information, and a daily stock summary; and outbound data flows from the ChemStock system to the Sales Representative including assigned stock details, stock alert summaries, and a delivery summary for the Sales Representative's current inventory responsibilities.

The Collector, with inbound data flows to the ChemStock system including delivery receipt confirmation details with photoproof and GPS location, and delivery confirmation details with photoproof and GPS location; and outbound data flows from the ChemStock system to the Collector including delivery assignment details for the Collector's field activities and for proper recording and traceability of all stock transfers handled by the Collector.

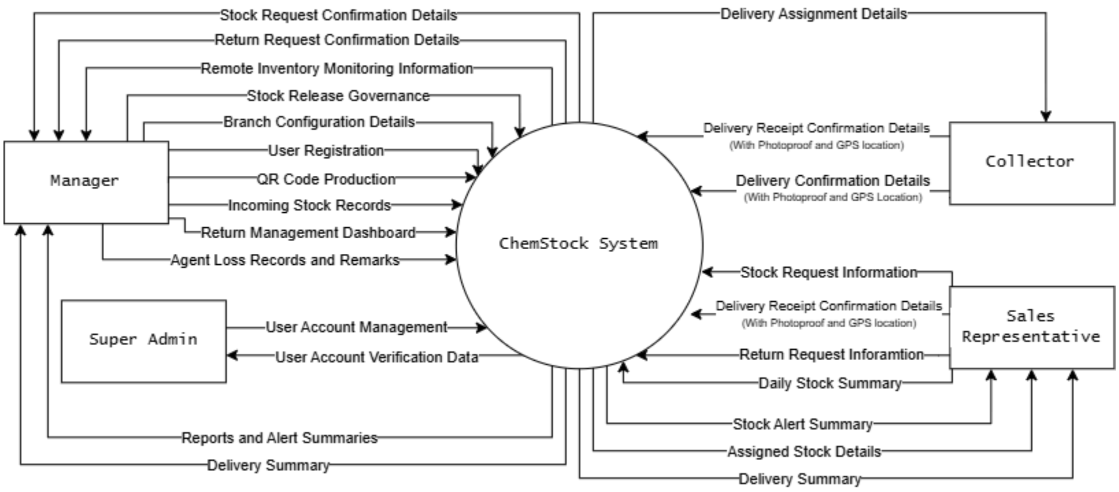


Figure 6. Context Diagram for ChemStock

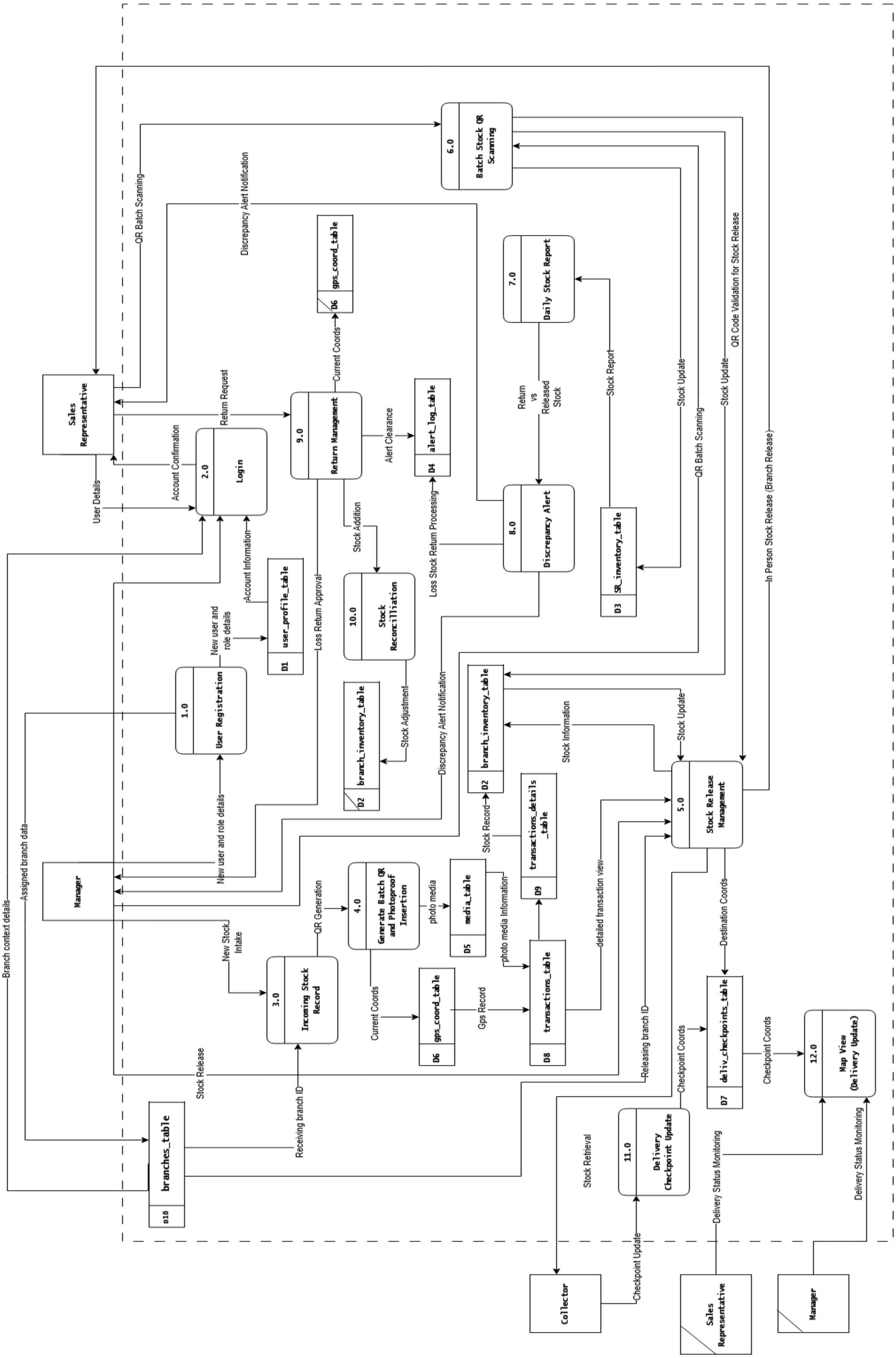
3.3.4 Data Flow Diagram

Figure 7 defines the Level 1 Data Flow Diagram (DFD) for the ChemStock system, mapping programmatic data transformations across twelve core processes, nine data stores, and three external entities: Manager, Sales Representative, and Collector. The operational workflow initializes at Process 1.0 (User Registration), which handles two distinct user provisioning workflows. First, the Super Admin generates unique system activation codes that are committed to the user_profiles_table and serve as cryptographic gatekeeping tokens for Branch Manager registration. Second, once registered using the activation code, the Branch Manager creates accounts for Sales Representatives and Collectors within their assigned branch, committing their credentials directly to the user_profiles_table. Crucially, this table maintains a foundational relational mapping to the branches_table data store to structurally isolate user permissions by branch parameters. The Super Admin does not directly create accounts for Sales Representatives or Collectors—this is exclusively the Branch Manager's responsibility. Authenticated sessions are subsequently governed by Process 2.0 (Login), which retrieves cached account records to verify all active system agents.

For inventory onboarding, Process 3.0 (Incoming Stock Record) ingests shipment manifests from the Manager and programmatically triggers Process 4.0 (Generate Batch QR and Photoproof Insertion). This pipeline captures localized telemetry parameters, writing real-time GPS coordinates to the gps_coord_table, inserting photographic media vectors into the media_table, and committing formal stock parameters directly into the branch_inventory_table. Process 5.0 (Stock Release Management) serves as the centralized core engine, routing both face-to-face and remote collector-driven shipments. It enforces a mandatory QR code validation sequence, pulling stock attributes from the branch_inventory_table while simultaneously streaming itemized historical vectors into the transaction_table and transaction_log data stores.

Field-level interactions continue at Process 6.0 (Batch Stock QR Scanning), which ingests batch matrix scans from the Sales Representative to execute synchronized balancing updates across both the SR_inventory_table and branch_inventory_table. Following deployment cycles, Process 7.0 (Daily Stock Report) reads real-time field balances from the SR_inventory_table to balance returns against original inventory releases, passing raw metrics to Process 8.0 (Discrepancy Alert). This alert engine writes active variances into the alert_log_table and pushes automated warning payloads back to the Sales Representative.

Reverse logistics are managed via Process 9.0 (Return Management), which processes inbound agent return manifests, queries verified coordinates from the gps_coord_table, purges cleared signals within the alert_log_table, and passes validated reconciliations to Process 10.0 (Stock Reconciliation) to dynamically adjust inventory counts inside the branch_inventory_table. Finally, field delivery workflows are updated by Process 11.0 (Delivery Checkpoint Update), which captures intermittent geolocation points from the Collector, registers destination parameters inside the deliv_checkpoints_table, and feeds spatial coordinates to Process 12.0 (Map View). This mapping loop renders real-time telemetry overlays, granting both the field-deployed Collector and the branch-level Manager comprehensive operational visibility over active transit operations.



Sales Representative
Manager
Collector
1.0 User Registration
2.0 Login
3.0 Incoming Stock Record
4.0 Generate Batch QR and Photoproof Insertion
5.0 Stock Release Management
6.0 Batch Stock QR Scanning
7.0 Daily Stock Report
8.0 Discrepancy Alert
9.0 Return Management
10.0 Stock Reconcilliation
11.0 Delivery Checkpoint Update
12.0 Map View (Delivery Update)
D1 user_profile_table
D2 branch_inventory_table
D3 SR_inventory_table
D4 alert_log_table
D5 media_table
D6 gps_coord_table
D7 deliv_checkpoints_table
D8 transactions_table
D9 transactions_details_table
D10 branches_table
User Details
Account Confirmation
Account Information
New user and role details
Return Request
QR Batch Scanning
Discrepancy Alert Notification
Current Coords
Alert Clearance
Stock Addition
Loss Return Approval
Loss Stock Return Processing
Return vs Released Stock
Stock Report
Stock Update
QR Code Validation for Stock Release
In Person Stock Release (Branch Release)
Stock Adjustment
Stock Information
Stock Record
photo media
photo media Information
QR Generation
New Stock Intake
Gps Record
detailed transaction view
Destination Coords
Checkpoint Coords
Releasing branch ID
Receiving branch ID
Stock Release
Stock Retrieval
Branch context details
Assigned branch data
Checkpoint Update
Delivery Status Monitoring

Figure 7. Data Flow Diagram Level 1 for ChemStock

3.3.5 Database Design

Figure 8 shows the Entity-Relationship Diagram (ERD) for the ChemStock system, defining the structure, constraints, and relational dependencies across ten core tables: user_profiles_table, branches_table, branch_inventory_table, transaction_table, transaction_details_table, gps_coord_table, media_table, alert_log_table, SR_inventory_table, and deliv_checkpoints_table.

Relational integrity initializes at the branches_table, which acts as the foundational master data registry for all physical branch locations. This table connects in a one-to-many relationship to the primary user_profiles_table, which stores each user's unique identifier, username, role classification, assigned branch foreign key, and cryptographically hashed password. By linking user identities to specific branches, the schema enforces role-based constraints and ensures that all downstream transactional entries and structural audits can be mapped back to authenticated personnel within isolated operational nodes. Concurrently, the branch_inventory_table handles bulk, branch-level asset tracking, documenting unique batch identifiers, alphanumeric product nomenclature, product expiration parameters, real-time warehouse quantities, and quick response (QR) code matrix tokens, while maintaining a foreign key link to the media_table for structural photo verification of inbound factory dispatches.

The core operational workflow is governed by the transaction_table, which serves as a centralized historical ledger recording every custody transfer event by indexing the managing supervisor, assigned field recipient, geolocation attributes, movement type, remarks strings, and offline synchronization status states. Each parent record in this ledger maintains a one-to-many relationship with the transaction_details_table, which isolates line-item specifics regarding product IDs and discrete quantities moving through the active distribution stream. Spatial and evidentiary validity layers are reinforced via the gps_coord_table and media_table, which index hardware-derived coordinates, secure media URLs, local device metadata, and immutable timestamps to validate the field chain of custody. Automated reconciliation gaps are covered by the alert_log_table, which writes real-time stock variance logs, capturing aggregate released units, accounted items, and calculated discrepancy differentials, by linking directly across the user_profiles_table and SR_inventory_table to instantly flag at-fault field agents. Finally, decentralized field allocations are dynamically updated inside the SR_inventory_table to balance individual agent inventory levels, while the deliv_checkpoints_table continuously registers transport coordinate logs and milestone timestamps to provide real-time delivery telemetry across active logistics networks.

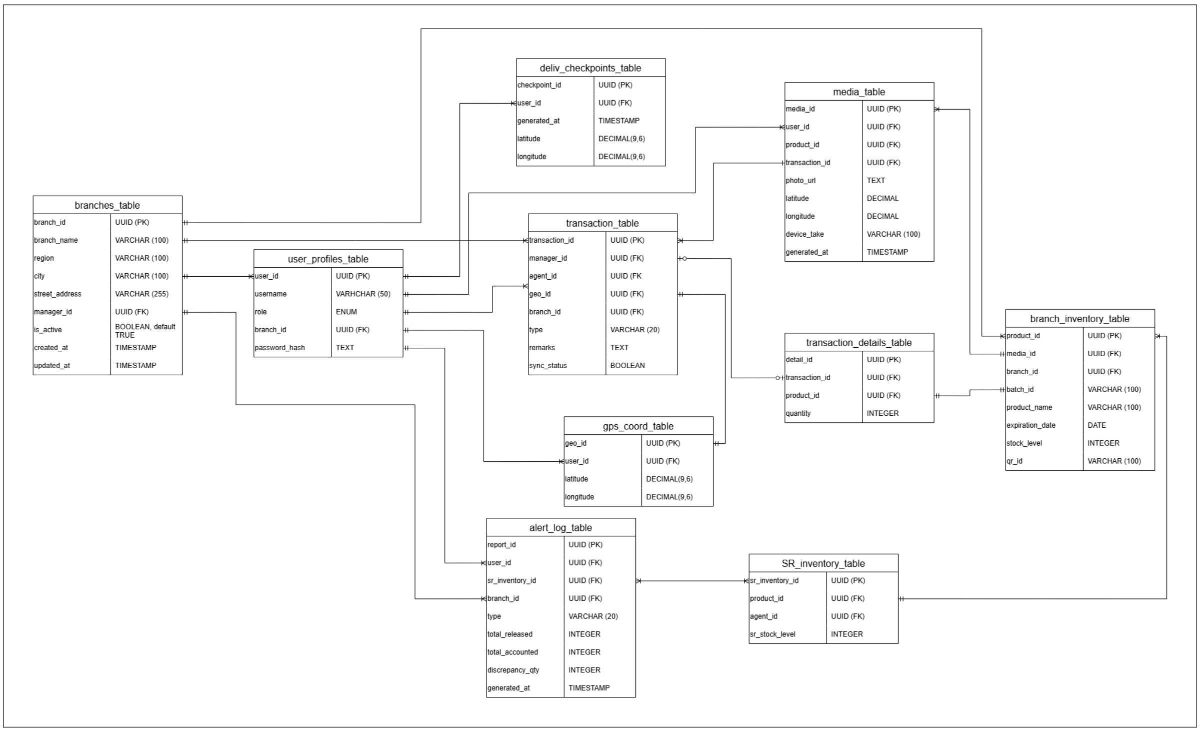


Figure 8. Database Design for ChemStock

3.4 Tech Stack

3.4.1 Software Requirements

Table 3 shows the Software requirements upon developing ChemStock

Table 3. Software requirements for ChemStock

Component	Technology	Description
Presentation Tier (Frontend)	**React Native (Expo) -** is a framework for building cross-platform mobile applications using JavaScript and React, with Expo providing tools and services that simplify development, testing, and deployment.	Researchers chose React Native (Expo) because it supports multiple APIs for features such as authentication, maps, QR code scanning and generation, camera access, notifications, and database integration, making development more efficient. Combined with its easy-to-learn language and streamlined tools, it helps accelerate development and ensures completion within the set timeframe.
Application Tier (Backend)	**Express JS (Node.js) -** is a backend mobile application framework used for building fast and scalable server-side applications and APIs using JavaScript.	Researchers chose Express.js (Node.js) as the backend framework because of its fast performance, scalability, and efficient handling of API requests and real-time data. Its JavaScript-based environment also allows smoother integration with React Native (Expo), making development more consistent and efficient.
Database (Data Tier)	**PostgreSQL Supabase** - is an open-source, object-relational database management system known for its reliability, data	The researchers chose PostgreSQL Supabase because it provides robust data integrity, ACID compliance, and real-time capabilities

	integrity, and support for advanced features such as real-time subscriptions and row-level security. PostgreSQL serves as the primary cloud database via Supabase, storing all centralized inventory records, user profiles, transaction logs, and geotagged chain of custody data while enabling real-time synchronization across all branch managers, sales representatives, and collectors.	essential for accurate inventory tracking. Its row-level security features allow fine-grained access control for different user roles (managers, sales representatives, collectors), while native real-time subscriptions enable instant discrepancy alerts and live inventory updates without manual page refreshes. Additionally, PostgreSQL's reliability ensures that critical transaction logs and geotagged chain of custody records remain consistent even during concurrent access by multiple users across different branches.
Media Storage (Data Tier)	**Supabase Bucket - i**s a cloud-based storage container within Supabase's Storage service, used to store and organize files like images, videos, and documents, with built-in security rules and integration with Supabase's authentication system."	Researchers chose Supabase buckets because they offer a simple, cost-effective, and secure way to store and manage user-generated files (images and documents) without needing to set up or maintain a separate cloud storage services. The tight integration with Supabase's authentication and row-level security allows researchers to enforce access controls easily, reducing development time and security risks, ideal for research-focused apps

		where budget and deployment speed are priorities.
Offline Sync Manager	**SQL Lite -** is a lightweight, embedded relational database engine that stores data locally on the device. In ChemStock, expo-sqlite enables offline functionality by temporarily holding transaction records and synchronizing them to the server once internet connectivity is restored.	The researchers chose SQLite (expo-sqlite) because it enables full offline functionality, allowing sales representatives to record transactions without internet connectivity. Its lightweight, serverless architecture ensures fast local data access and seamless synchronization with Supabase once the connection is restored, which is essential for ChemStock's distributed sales operations in areas with unreliable internet.
QR Code Generation Library	**Expo-Barcode-Generator or React-Native-qrcode-svg -** is a library that generates QR codes and barcodes directly within Expo-based applications without requiring native dependencies. It enables the system to create scannable QR codes for each product batch, allowing managers to assign unique identifiers to inventory items for efficient tracking and FIFO compliance.	The researchers chose React Native QRCode SVG because it generates scannable QR codes directly within the mobile application without requiring external API calls or internet connectivity. This ensures that managers can create batch QR codes on demand, even when offline, and eliminates dependency on third-party QR generation services, reducing costs and improving data privacy.
Version Control	**Git / GitHub -** is a	Researchers chose

	web-based platform used for version control and collaborative software development, allowing developers to store, manage, and track changes in their code using Git.	GitHub because it provides reliable version control, enables collaborative development among team members, and helps track and manage code changes efficiently throughout the development of the mobile application.
UI/UX Design Tool	**Figma** - is a cloud-based design and prototyping tool used for creating user interface (UI) and user experience (UX) designs, allowing real-time collaboration among team members.	Researchers chose Figma because it allows efficient UI/UX design and real-time collaboration, enabling the team to create, review, and refine the application interface quickly and consistently before development.
Diagram Design Tool	**Draw.io** - is a free, web-based diagramming tool used for creating flowcharts, system architectures, ER diagrams, and other visual models for planning and documentation.	Researchers chose draw.io (diagrams.net) because it provides a simple and flexible way to create clear system diagrams and flowcharts, helping the team visualize system architecture, database relationships, and workflows for better planning and documentation.

3.4.2 Hardware Requirements

The table 4 shows the Hardware Requirements upon developing ChemStock

Table 4. Hardware Requirements for ChemStock

Hardware Component	Minimum Specification	Operational Justification
Operating System	Android 10 +	Required for compatibility with Expo managed workflow, modern camera APIs (expo-camera), and location services (expo-location) used in the app
Processor	Quad-core 1.8 GHz or equivalent	Sufficient to handle QR scanning, photo capture, GPS tracking, and local database operations (expo-sqlite) without performance lag
Memory (RAM)	2 GB	Minimum needed to run React Native (Expo) apps smoothly while multitasking with camera, GPS, and synchronization processes
Storage	32 GB internal storage (with at least 500 MB free)	Required for storing app binaries, offline SQLite database, QR batch data, and captured photo evidence (media_table) before cloud sync
Network	Mobile Data (4g/LTE) or Wifi	A persistent internet connection is required to synchronize with the Express JS backend. In offline scenarios, activities are queued locally and automatically synced once connectivity is restored."
Camera	Rear camera with autofocus (5 MP or	Essential for QR code scanning (expo-camera)

	higher)	and photo handover documentation
GPS / Location	GPS / A-GPS support	Required for geotagging functionality (expo-location) to capture coordinates during stock release and receipt confirmation
Sensors	Accelerometer, proximity sensor	Used by Expo libraries for device orientation and camera control during scanning
Display	5.5 inches (HD+)	Ensures inventory lists and QR scan previews are clearly readable, reducing data entry errors.

3.5 Research Instrument

To evaluate the effectiveness of ChemStock compared to the existing manual inventory management system, the researchers employed a mixed-methods approach combining qualitative and quantitative research instruments. The evaluation focused on measuring improvements against the five specific problems identified in section 1.2.2: late notice of stock losses (Problem A), absence of real-time inventory visibility (Problem B), accountability gaps and unverifiable chain of custody (Problem C), manual recording of errors and cumbersome inventory counting (Problem D), and lack of evidence for physical stock handovers (Problem E). The following instruments are to be administered after a three week pilot implementation (first week is allocated for user deployment and training, followed by two weeks of active evaluation) period across both the Cagayan de Oro and Butuan Branches.

3.5.1 User Acceptance Testing (UAT) Survey Questionnaire

The researchers developed a structured survey questionnaire to measure user satisfaction, perceived usefulness, and ease of use of ChemStock mobile application. The questionnaire is designed using a five-point likert scale (1 = Strongly Disagree, 5 = Strongly Agree) and consisted of twelve core statements distributed across six evaluation categories: inventory visibility, QR scanning efficiency, photo handover accountability, geotagging and chain of custody, loss tracking and alerts, and overall satisfaction. Two-open ended questions were included at the end of the survey to capture qualitative feedback regarding user preferences, challenges encountered, and suggestions for improvement. The survey will be administered to all 39 staff members across both branches, including branch managers, sales representatives, and collectors.

Respondent Profile:

☐ Branch Manager

☐ Sales Representative

☐ Collector

Branch: ☐ Cagayan de Oro ☐ Butuan

5 = Strongly Agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly Disagree

#	Statement	5	4	3	2	1
A. Inventory Visibility (Problem B)						
A1	I can view current stock levels without being physically present at the branch office.	☐	☐	☐	☐	☐
A2	The dashboard provides me with accurate and up-to-date inventory information.	☐	☐	☐	☐	☐
B. QR Scanning Efficiency (Problem D)						
B1	QR scanning is faster than manually counting and recording stocks.	☐	☐	☐	☐	☐
B2	Using QR codes reduces my data entry errors compared to paper-based recording.	☐	☐	☐	☐	☐
C. Photo Handover & Accountability (Problem C)						
C1	The photo handover feature helps resolve disputes about stock deliveries.	☐	☐	☐	☐	☐
C2	Having timestamped photographic evidence makes me more confident during stock transfers.	☐	☐	☐	☐	☐
D. Geotagging & Chain of Custody (Problem C)						
D1	GPS location capture during stock release/receipt provides verifiable proof of transactions.	☐	☐	☐	☐	☐
D2	The geotagged records protect me from false accusations of non-receipt.	☐	☐	☐	☐	☐
E. Loss Tracking & Alerts (Problem A)						
E1	The system helps me identify stock discrepancies faster than the manual method.	☐	☐	☐	☐	☐
E2	Automated alerts notify me promptly when a stock discrepancy is detected.	☐	☐	☐	☐	☐
F. Overall Satisfaction						
F1	ChemStock is easier to use than the previous paper-based system.	☐	☐	☐	☐	☐
F2	I would recommend ChemStock to other branches of Cospachem.	☐	☐	☐	☐	☐

Open Ended Questions:

1. What do you like most about ChemStock compared to the manual system?
2. What challenges or difficulties have you encountered while using ChemStock?
3. What improvements would you suggest for future versions?

Figure 9. UAT Survey Questionnaire

3.5.2 Time-Motion Study Log

To quantitatively compare task completion times between manual and ChemStock-enabled processes, the researchers will conduct a time-motion study. Time-motion methodology is a validated approach for evaluating the efficiency of digital interventions. Previous research has successfully used time-motion logs to prove that digital recording systems significantly reduce the time users spend on information management compared to traditional paper-based documentation (Venkateswaran et al., 2022). Five critical inventory tasks were measured: recording incoming stock delivery, releasing stocks to a sales representative, confirming receipt of stock delivery, generating a weekly inventory report, and performing a full inventory count. For each task, the researchers will record the start time and end time (in minutes) for both the manual method (baseline) and the ChemStock method. Measurements are taken during the second and third week of the pilot implementation to account for initial learning curves. The time-motion study log will record the user role, branch location, and any observational notes relevant to task performance.

Time-Motion Study Log

Observer Name: ________________________ **Date:**

Instructions: Record the time (minutes) taken to complete each task using the assigned method

Task ID	Task Description	Method	Start Time	End Time	Total Time (min)
T1	Recording incoming stock delivery	Manual	:	:	=
T1	Recording incoming stock delivery	ChemStock	:	:	=
T2	Releasing stocks to a sales representative	Manual	:	:	=
T2	Releasing stocks to a sales representative	ChemStock	:	:	=
T3	Confirming receipt of stock delivery	Manual	:	:	=
T3	Confirming receipt of stock delivery	ChemStock	:	:	=
T4	Generating weekly inventory report	Manual	:	:	=
T4	Generating weekly inventory report	ChemStock	:	:	=
T5	Performing full inventory count	Manual	:	:	=
T5	Performing full inventory count	ChemStock	:	:	=

Figure 10. Time-Motion Study Log

3.5.3 Error and Discrepancy Log Analysis

To measure the reduction in stock discrepancies and loss detection timing, the researchers will implement a daily error and discrepancy log. The log captured the following data for each discrepancy identified during the three-week pilot period: transaction identification number, date of occurrence, product name, expected quantity, actual quantity, discrepancy magnitude (positive or negative), who detected the discrepancy, time elapsed between transaction and detection, and identified root cause (if determinable). Daily summary metrics will be computed, including total number of discrepancies, total stock units lost, average time to detect discrepancies, and number of unresolved disputes. These metrics are then compared against the manual baseline 2 week data which will be obtained before implementation occurs.

Error and Discrepancy Log Analysis

Branch: ☐ Cagayan de Oro ☐ Butuan Week #: Date Range:

Transaction ID	Date	Product	Expected Qty	Actual Qty	Discrepancy (+ / -)	Detected By	Time to Detect	Root Cause (if identified)

Weekly Summary:

Metric	This Week	Cumulative (since implementation)
Total number of discrepancies		
Total stock units lost		
Average time to detect discrepancy (hours)		
Number of unresolved disputes		

Comparison with Manual Baseline:

Metric	Manual Baseline (2 week avg)	ChemStock (2 - week pilot avg)
Annual Loss		
Detection Time		
Recovery rate		

Figure 11. Error and Discrepancy Log Analysis

3.5.4 System Usability Scale (SUS)

To obtain a standardized, validated measure of overall system usability, the researchers will administer the System Usability Scale (SUS) developed by Brooke (1996). The SUS consists of ten statements rated on a five-point Likert scale, alternating between positive and negative phrasing to reduce response bias. The ten statements assess frequency of system use, perceived complexity, ease of use, need for technical support, feature integration, inconsistency, learnability, cumbersomeness, user confidence, and prerequisite learning burden. The SUS score will be calculated using the standard formula: for odd-numbered items (1,3,5,7,9), the score contribution is the response minus one; for even-numbered items (2,4,6,8,10), the score contribution is five minus the response. The sum of all contributions is multiplied by 2.5 to produce a final SUS score ranging from 0 to 100. Scores will be interpreted using the standard adjective rating scale: 80-100 (Excellent), 68-79 (Good), 50-67 (OK/Marginal), and 0-49 (Poor). The SUS will be administered alongside the UAT survey to all 39 participants during the third week of the pilot implementation.

System Usability Scale (SUS)

		Strongly Disagree		Neutral		Strongly Agree
1	I would like to use ChemStock frequently.	1	2	3	4	5
2	I found ChemStock unnecessarily complex.	1	2	3	4	5
3	I thought ChemStock was easy to use.	1	2	3	4	5
4	I would need technical support to use ChemStock.	1	2	3	4	5
5	I found the various features (QR, GPS, photo) were well integrated.	1	2	3	4	5
6	I thought there was too much inconsistency in ChemStock.	1	2	3	4	5
7	I would imagine most people would learn ChemStock very quickly.	1	2	3	4	5
8	I found ChemStock very cumbersome to use.	1	2	3	4	5
9	I felt very confident using ChemStock.	1	2	3	4	5
10	I needed to learn a lot before I could use ChemStock.	1	2	3	4	5

Figure 12. System Usability Scale

3.5.5 Pre Implementation vs Post Implementation Comparison Checklist

To synthesize all evaluation data into an overall effectiveness assessment, the researchers developed a comparison checklist that directly mapped each specific problem from Section 1.2.2 to measurable indicators. For each problem, the checklist documented the manual baseline (obtained from the initial 2-week before implementation), the ChemStock observed result (measured during the three-week pilot), a categorical improvement rating (Yes, Partial, or No), and the specific evidence source supporting the rating. The eight indicators evaluated included: detection time for stock losses, annual stock loss quantity, manager reimbursement cost, real-time visibility availability, verifiable chain of custody existence, frequency of recording errors, inventory counting time, and availability of photographic transaction proof. The checklist also included an overall effectiveness rating (Highly Effective, Moderately Effective, Slightly Effective, or Not Effective) based on the evaluating researcher's professional judgment after reviewing all collected data.

Effectiveness Evaluation Checklist: Manual vs ChemStock

Evaluator Name: ____________ **Branch:** ☐ CDO ☐ Butuan

Evaluation Date: ____________ **Weeks Since Implementation:** _____

Specific Problem	Manual Baseline	ChemStock	Improvement?
A. Late Notice of Stock Losses	Detection Time:	Detection Time:	☐ Yes ☐ Partial ☐ No
A. Annual Stock Losses	____ units (last 2 weeks)	____ units (last 2 weeks)	☐ Yes ☐ Partial ☐ No
A. Manager Reimbursement	₱_____ (projected)	₱_____ (projected)	☐ Yes ☐ Partial ☐ No
B. Inventory Visibility	No remote viewing; manager takes photos of paper records	Dashboard accessible remotely	☐ Yes ☐ Partial ☐ No
C. Verifiable Chain of Custody	No systematic record; disputes unresolved	Geo-tagged + photo proof	☐ Yes ☐ Partial ☐ No
D. Manual Recording Errors	Overlooked items	____ errors recorded in log	☐ Yes ☐ Partial ☐ No
D. Inventory Counting Time	1-2 hours per delivery	____ minutes per delivery	☐ Yes ☐ Partial ☐ No
E. Proof of Transactions (Stock Delivery Release)	No photographic or digital evidence	Timestamped photo per handover	☐ Yes ☐ Partial ☐ No

Figure 13. Effectiveness Evaluation Checklist

3.5.6 Data Collection Timeline

The data collection procedure will follow a structured three-week timeline. During Week -2 to -1, which serves as the pre-implementation phase, the researchers will conduct baseline data collection by obtaining manual comparison data from the branch managers through the initial structured interview. Week 1 will be dedicated to system deployment and user training, during which no evaluation instruments will be administered as users are being oriented to the ChemStock mobile application. In Week 2, the first evaluation cycle will be commenced, with the researchers administering the Time-Motion Study Log and initiating the daily Error and Discrepancy Log. Finally, Week 3 will serve as the final evaluation cycle, during which the researchers will administer the second round of the Time-Motion Study Log, collect the final entries of the Error and Discrepancy Log, and distribute the User Acceptance Testing (UAT) Survey and the System Usability Scale (SUS) Questionnaire to all 39 participants.

3.6 Ethical Considerations

The development of ChemStock necessitates adherence to ethical principles commensurate with the handling of sensitive transaction data, location information, and photographic evidence of inventory movements. The following considerations governed all phases of the research.

Informed Consent and Voluntary Participation

All research participants, including branch managers, sales representatives, and collectors from the Cagayan de Oro and Butuan branches (total 39 staff), were engaged with full disclosure of the study's purpose, scope, and their role therein. Participation was strictly voluntary, and respondents retained the right to withdraw at any point without consequence. Consent was obtained and documented prior to any data collection activity, in accordance with established ethical standards for social research involving human subjects. The informed consent form explicitly stated that the system captures GPS coordinates and photo documentation during stock release and receipt activities, and that no continuous background tracking occurs.

Data Privacy and Confidentiality

The app collects inventory levels, agent loss records, and photo proof of stock handovers. We used strict data protection rules. GPS coordinates are captured only at stock release and receipt confirmation. The app never tracks users in the background. Photo evidence is stored in Supabase Bucket storage with signed URLs that expire after 60 seconds. Only managers can access these photos. User names and branch locations are stored in the database with row-level security turned on. In published results, we will show only totals and role names like "Sales Representative A." We will not use real names. The system does not share transaction logs with outside people.

Protection of Participants from Unfair Accountability

Recognizing that sales representatives operate in the field with limited oversight, the system was designed with deliberate safeguards against false accusations. The mandatory photo handover requirement includes timestamp and GPS capture at the point of receipt confirmation, providing objective evidence that protects sales representatives from being held accountable for losses that occurred before they received the stock. Similarly, the dual-location capture (release point and receipt point) establishes a verifiable chain of custody that can exonerate an agent if the discrepancy occurred during collector transit. The client (branch management) agreed that discrepancy alerts require manager investigation before any financial liability is assigned, consistent with company policy that annual reimbursements only follow verified losses.

Fairness in Algorithmic Decision-Making

The automated reconciliation engine compares released stock quantities against reported sales and returns to generate discrepancy alerts. To prevent false positives, the system uses exact quantity matching with no weighting or bias. Discrepancy alerts are presented as notifications requiring manager investigation, not as automatic determinations of agent fault. The system explicitly does not incorporate performance metrics (e.g., historical loss rates) into alert generation, ensuring each discrepancy is evaluated independently based solely on transaction data.

Transparency and Honest Representation of Limitations

The research team acknowledges, and has disclosed within the study's scope and limitations (Section 1.5), the following constraints: GPS accuracy varies by device quality and environmental factors; the system cannot physically prevent loss or theft, only document it; historical inventory data exists only in paper format and will not be migrated; the system is tested only on Cagayan de Oro and Butuan branches (39 staff), not all 130+ branches nationwide. Furthermore, the pilot evaluation spans three weeks, which is insufficient to measure detection time for losses that would otherwise only surface during annual audits. This limitation is explicitly stated in the evaluation methodology.

Legal and Institutional Compliance

ChemStock will be designed in full cognizance of Philippine data privacy regulations under Republic Act No. 10173 (Data Privacy Act of 2012). The system does not collect sensitive personal information as defined by the Act (e.g., health, religion, political affiliations). Location data is collected only with explicit user consent per app permissions, and users may disable GPS at any time (though this prevents transaction completion). The company (A and Aimee Laboratories / Cospachem Products) retains ownership of all transaction data. No government database integration is required. The branch managers of Cagayan de Oro and Butuan branches validated the data collection procedures prior to pilot implementation.

3.7 Gantt Chart

Figures 14 and 15 presents the project's Gantt chart, mapping out a sequential 23-week software development life-cycle tailored for capstone project execution. The timeline is partitioned into two primary operational phases to manage milestones effectively. The initial 21-week block focuses on executing deliverables 1 through 10, encompassing foundational environment setup, UI/UX wireframing, relational database schema architecture, and the sequential development of core system modules. This phase culminates in a critical midterm evaluation and production of demo, establishing an iterative feedback loop for system optimization and refinement. Conversely, the final block spanning weeks 22 through 23 is strictly dedicated to deliverables 11 and 12. This period isolates high-level quality assurance testing, exhaustive documentation finalization, and simulated mock defenses to mitigate operational gaps and guarantee absolute system validation preceding the final capstone defense.

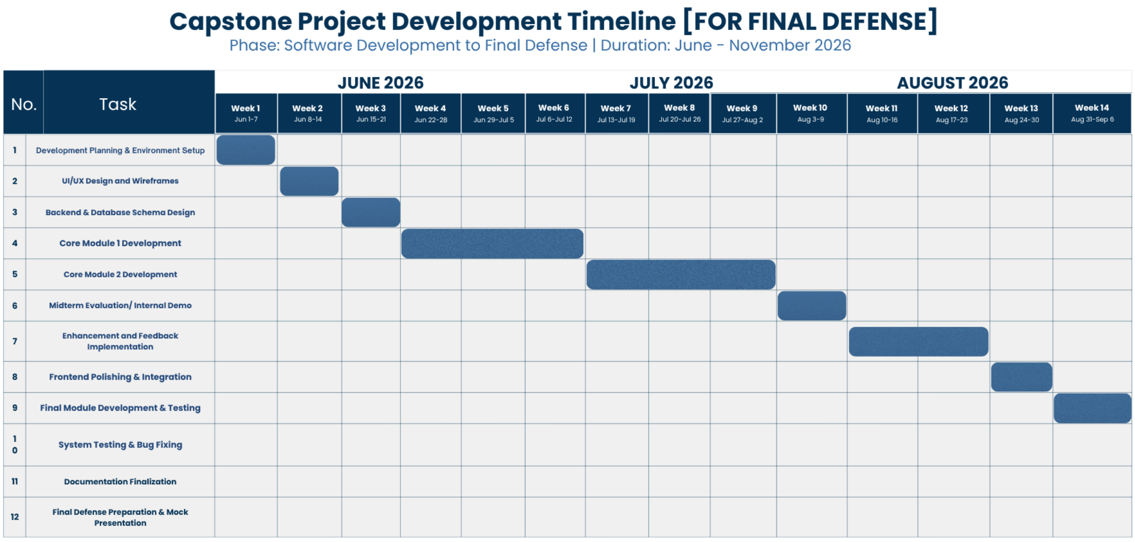


Figure 14. Gantt Chart for ChemStock (1)

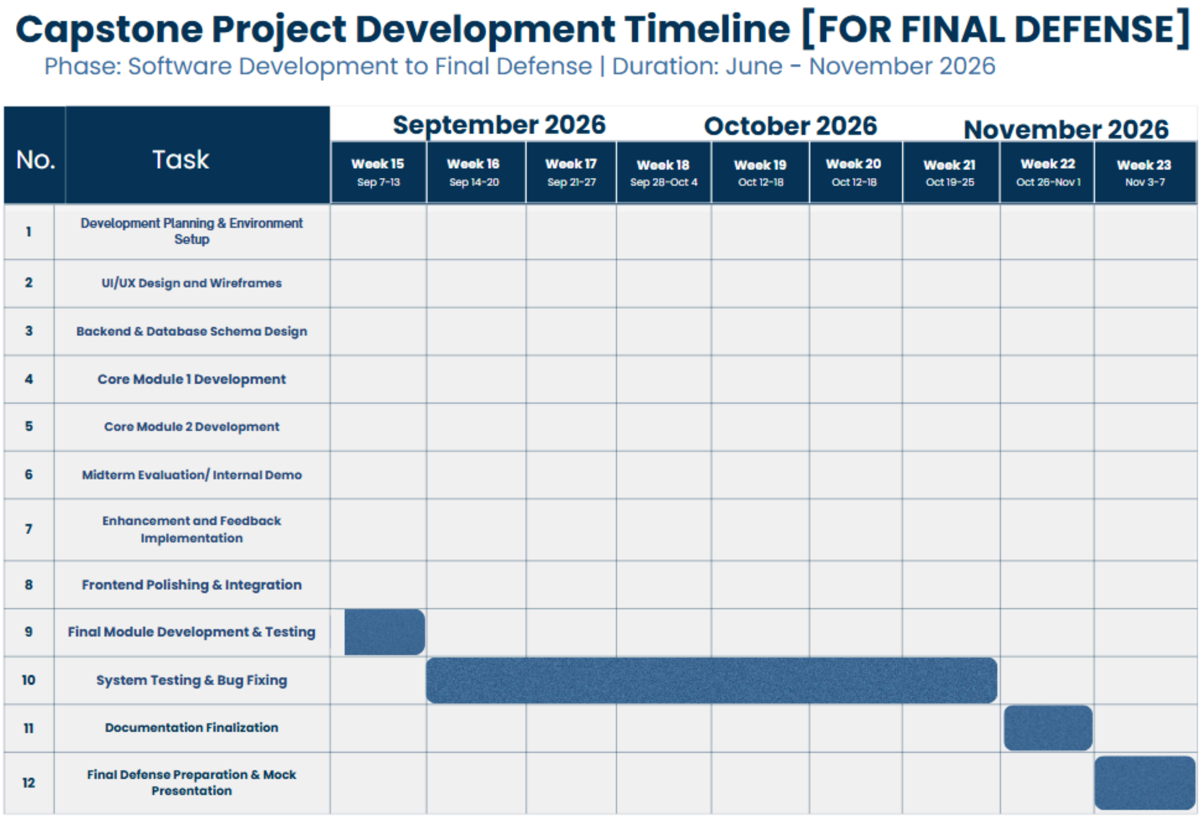


Figure 15. Gantt Chart for ChemStock (2)

3.8 System Prototype

To operationalize the system requirements defined in the Task-Technology Fit (TTF) framework and ensure seamless alignment with the data architectural design, a high-fidelity prototype of the ChemStock mobile platform was developed using Figma. The presentation tier is structured specifically to address the distinct user roles, namely Branch Managers, Sales Representatives, and Collectors, while executing core system processes both in online and localized offline states.

3.8.1 Manager POV System Prototype

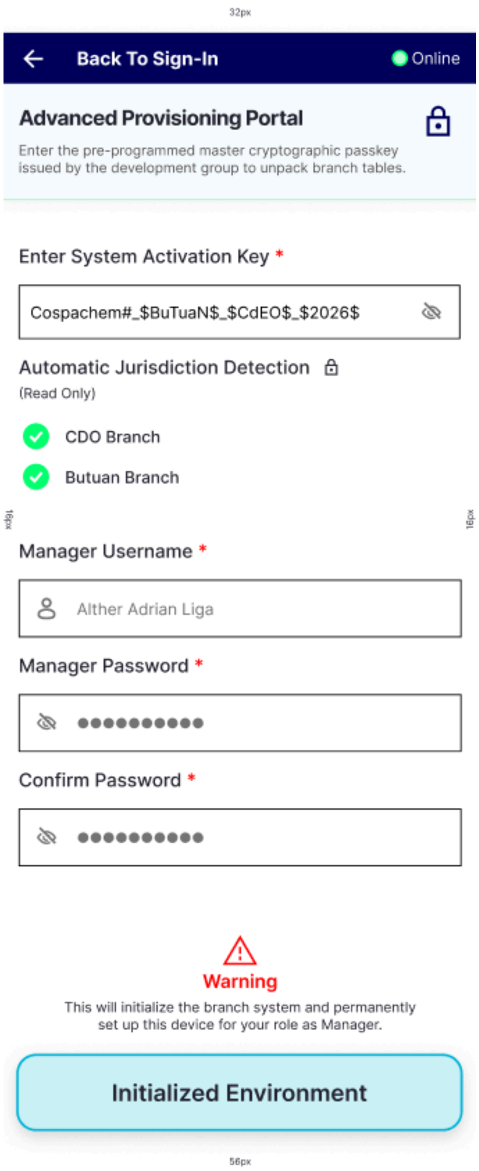


Figure 16. Manager Advanced Provisioning Portal Interface

Figure 16 captures the advanced cryptographic provisioning portal interface configured for manager account registration. Within this interface, account generation privileges are strictly isolated to the branch manager user tier, requiring a developer-issued System Activation Key to execute the verification sequence. This high-security activation key acts as a gatekeeping token that programmatically parses pre-set developer records to automatically detect and bind the manager's profile to their specific geographical jurisdiction and branch schema. To complete registration, the interface requires the input username and a secure password. Upon successful validation and database commitment within the user_profiles_table, the system instantiates the new profile under a standardized, sequential “Manager #” system identifier. Detailed identity information, including the user's actual name and custom profile metadata, are locked during initialization and can only be modified post-registration via the dedicated settings application module.

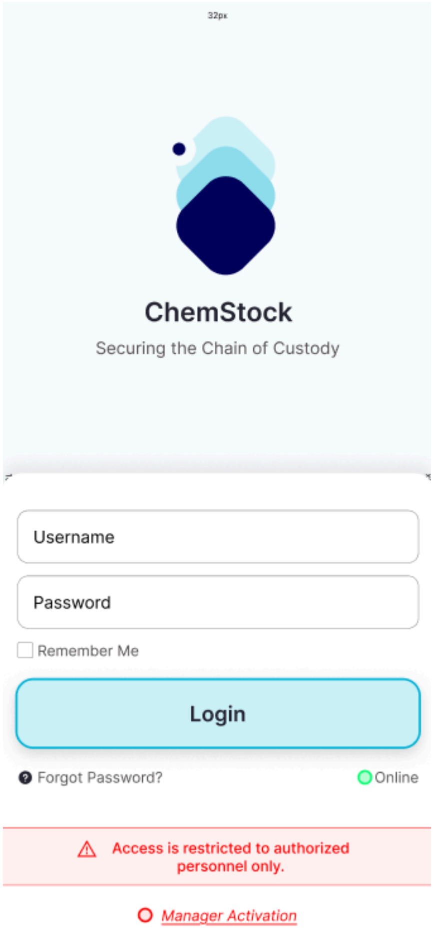


Figure 17. ChemStock Login Page

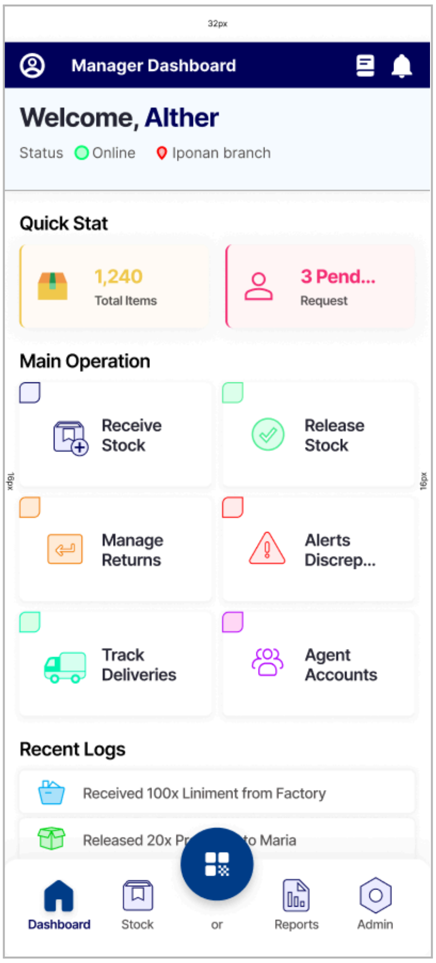


Figure 18. Manager Dashboard

Figure 17 illustrates the centralized system authentication interface connected to the user_profiles_table database schema. Operating under strict Role-Based Access Control (RBAC) protocols, the interface dynamically parses user inputs to validate cryptographic credentials against hashed password attributes. Upon a successful cryptographic handshake, the system session router enforces state-driven redirection, delivering role-specific authorization tokens to the presentation layer based on the user's documented classification as a Manager, Sales Representative, or Collector. Moreover, the option “Manager Activation” is the action that transfers users to figure 16 or the dedicated cryptographic provisioning portal interface exclusive for manager user tier registration.

Figure 18 captures the primary administrative dashboard interface configured exclusively for the branch manager user tier, establishing the central control layer for localized inventory management and personnel oversight. The presentation layer features a grid of core execution nodes that streamline the branch's daily workflow, including stock reception, stock release, return management, discrepancy monitoring, active delivery tracking, and field agent account administration. Positioned at the bottom of the interface, a persistent navigation bar provides direct access to deep-dive sub-systems: the “Stock” node transitions the user to the branch_inventory_table ledger, the “Reports” utility initiates automated monthly summary compilation, and the “Admin” node reveals system configuration and security settings.

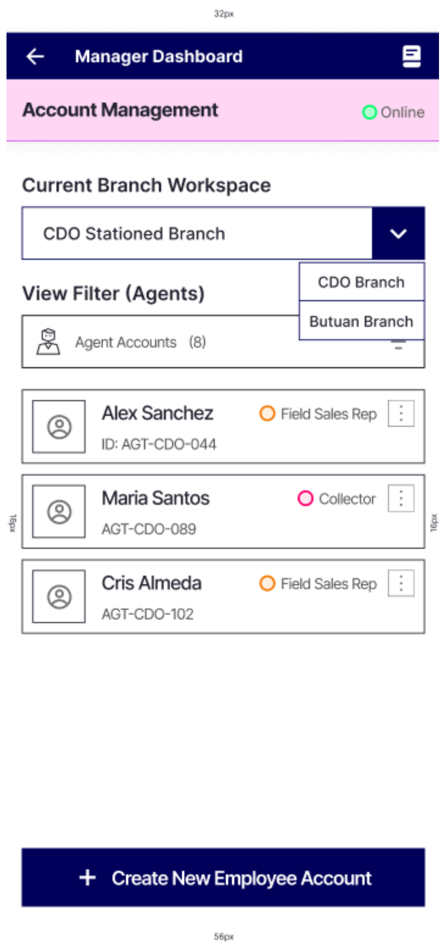


Figure 19. Manager Account Management Interface

Figure 19 captures the identity and account management interface configured exclusively for the manager user tier, serving as the localized administrative control node for branch-level provisioning. Within this interface, authorized managers can programmatically instantiate unique user profiles for field sales representatives and collectors, dynamically binding each credential set to a verified branch location block. This provisioning pipeline strictly enforces the system's underlying Role-Based Access Control (RBAC) rules matrix; by isolating account creation privileges entirely within the manager tier, the architecture prevents unauthorized profile registration and limits ledger vulnerabilities. Created account credentials are subsequently distributed securely to target personnel, establishing an absolute, closed-loop tracking mechanism for branch staff operations.

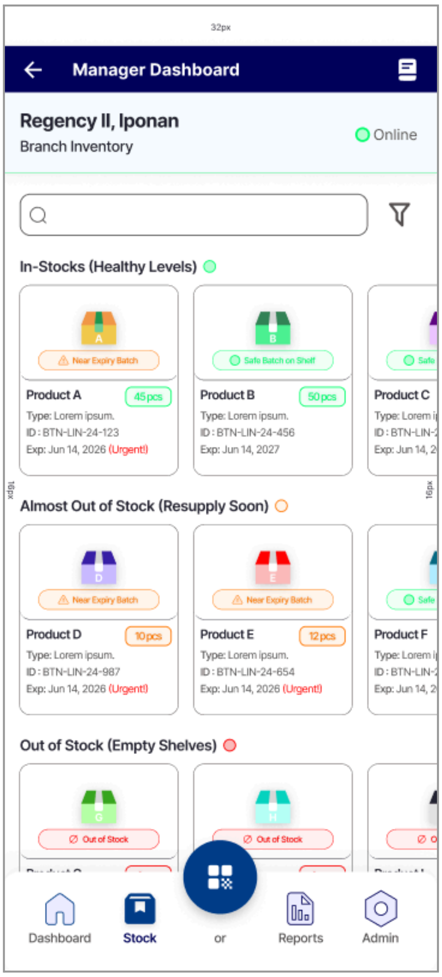


Figure 20. Manager Branch Inventory Record Interface

Figure 20 captures the branch inventory record interface configured for the manager user tier, establishing an analytical dashboard layer for stock oversight. Within this control screen, the interface displays current stock assets actively held in the branch's custody by pulling the data directly from the branch_inventory_table. The presentation layer surfaces precise metric indicators to categorize inventory health dynamically into three distinct operational tiers: healthy stock levels, near-depletion thresholds, and out-of-stock variances. Furthermore, the interface explicitly references product expiration dates and highlights critical items requiring immediate dispatch. This cross-referencing logic provides managers with actionable predictive visibility, enabling optimized, data-driven stock allocation and distribution workflows.

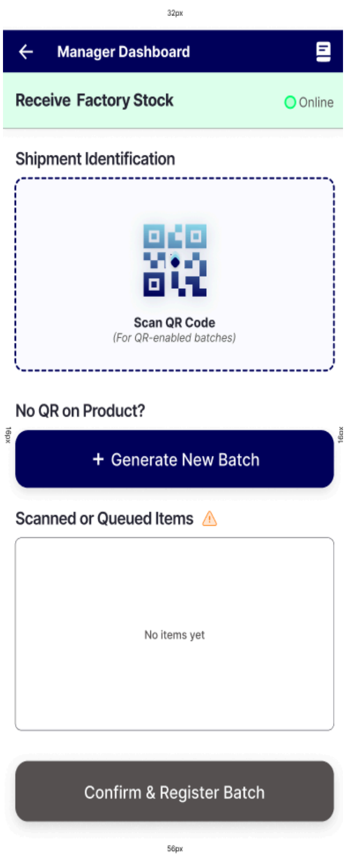


Figure 21

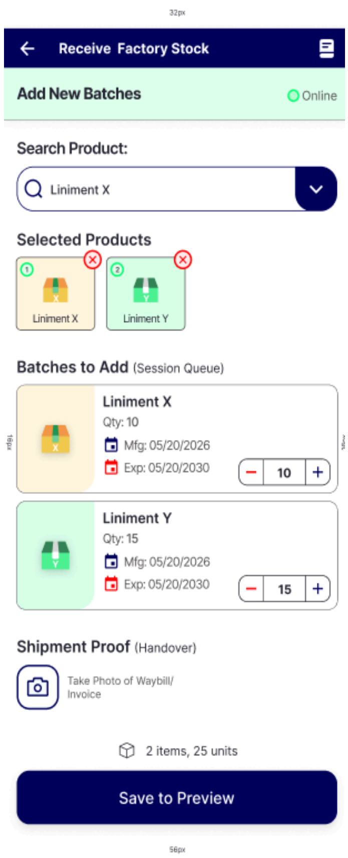


Figure 22

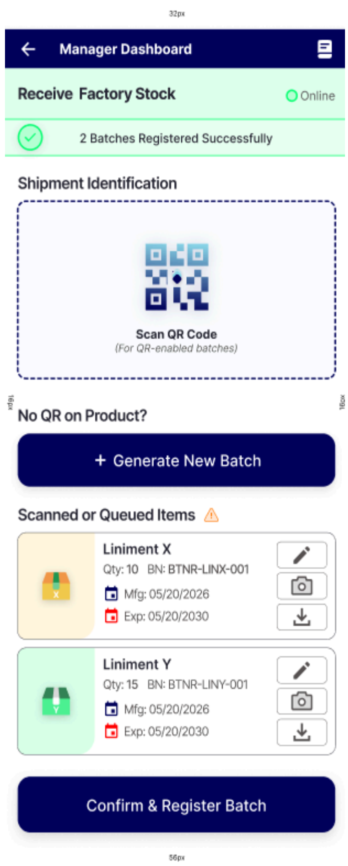


Figure 23

Figure 21 - 23. Manager Receive Stock and QR Generation Interface

Figure 21 captures the structural layout of the inventory receiving module executed under the Branch Manager user tier. This interface serves as the primary data intake channel, allowing the administrative user to select between two distinct choices: utilizing the integrated hardware scanner to decode pre-existing QR codes, or choosing “Generate New Batch” to generate a new QR code batch dynamically bound to specific product models. Upon a successful hardware scanning, the application layer updates the session state to display the parsed inventory rows under the "Scanned or Queued Items" panel. This localized tracking environment provides a temporary visual buffer for itemized verification prior to committing the final stock modifications to the primary database, a process explored comprehensively in Figure 23.

Figure 22 captures the creation of new QR batches in cases when there is no available scannable QR that the manager can utilize to immediately scan items. Inventory rows are categorized by specific product lines and sorted by expiration parameters to support programmatic compliance with the company's First-In, First-Out (FIFO) structural policy. Interactive filter utilities allow managers to query specific batch identifiers and cross-examine remaining quantities without requiring a physical presence at the office.

Figure 23, on the other hand, captures the presentation tier state of the inventory receiving module immediately following a successful matrix code execution sequence. The interface dynamically parses and renders the encoded data packets extracted from the designated QR codes, reflecting metadata schemas such as batch identifiers, product classifications, and primary metrics. Structural modification tools are provided to ensure data accuracy before ledger entry: an alphanumeric input utility (represented by a pencil icon) allows for real-time quantity or description overrides; a native camera interface bridge (represented by a camera icon) allows the manager to capture and attach photographic evidence directly to the media_table storage container; and a deployment utility (represented by a download icon) allows the generated matrix codes to be exported locally for physical distribution.

Upon triggering the confirmation sequence, the application layer runs structural validation rules and commits the cached dataset, writing all currently scanned and queued item arrays into persistent storage within the branch_inventory_table relational database schema.

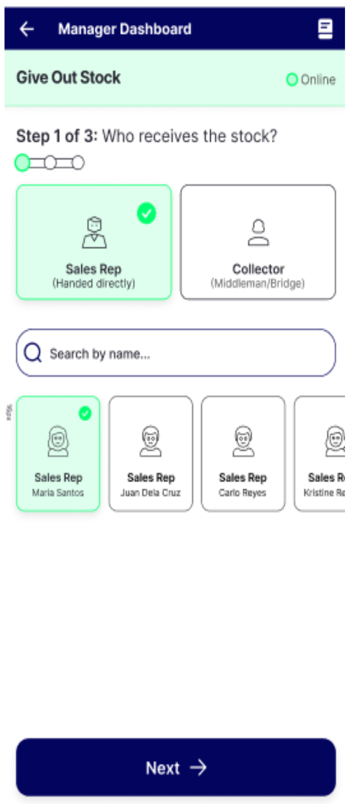


Figure 24

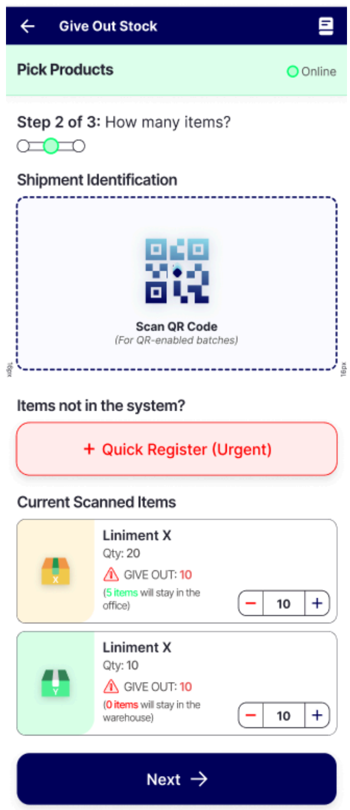


Figure 25

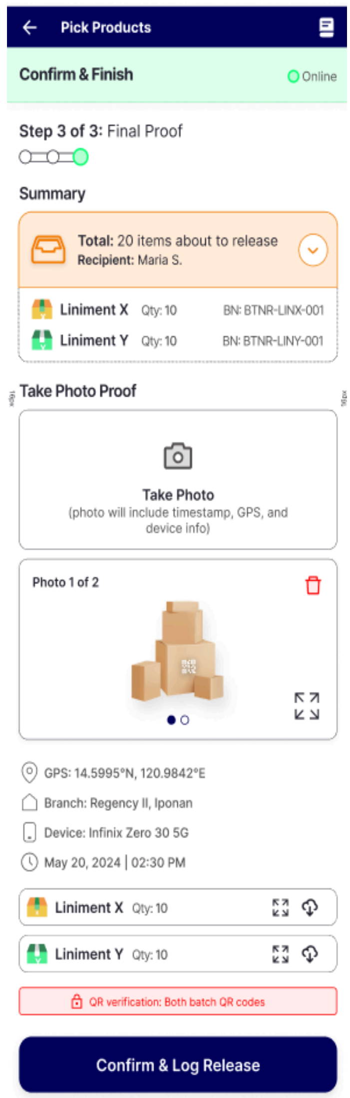


Figure 26

Figure 24 - 26. Manager Direct Stock Release for Sales Reps Interface

Figure 24 captures the interface for the distributed stock release module, representing the frontend framework for Process 5.0 (Stock Release

Management). The interface offers the administrative user a structural method selection mechanism to accommodate different distribution channels: "Handed Directly" for immediate, face-to-face transactions executed at the physical branch office, or "Middleman" for remote routing pipelines where stock is dispatched via a collector acting as an active courier intermediate. In the scenario illustrated, the "Handed Directly" parameter has been selected, bypassing intermediary delivery checkpoints. This mode allows the branch manager to map product allocations directly by selecting a designated sales representative account. The personnel records displayed within this selector component are dynamically retrieved through relational queries from the user_profiles_table data store, ensuring strict role validation before the system creates a formal transaction record.

Figure 25 captures the second step interface for the in-person stock release sequence. Through this interface, the branch manager can utilize the integrated mobile scanning hardware to decode pre-existing QR codes, dynamically retrieving the corresponding dataset from the branch_inventory_table. To ensure operational continuity during urgent periods where product batches lack pre-registered QR codes, the system incorporates a "Quick Register (Urgent)" protocol. This serves as an immediate navigational bridge back to the QR batch generation interface detailed in Figure 22, minimizing workflow friction. Once code verification is successfully executed, the system updates the session state, rendering the itemized product arrays within the "Current Scanned Items" buffer to allow for validation prior to ledger finalization.

Figure 26 captures the manager's Stock Release Confirmation interface. Through this interface, the system aggregates the transaction metadata to display a comprehensive summary of all scanned items slated for distribution, while providing localized utilities to export the respective batch QR codes. To mitigate accountability gaps, the workflow enforces a mandatory "Take Photo Proof" security protocol. This feature bridges directly to the device's native camera API to prevent the fraudulent importation of historical files from the local storage gallery, ensuring that only real-time photographic evidence is accepted. Upon image capture, the system dynamically embeds unalterable contextual telemetry into the ledger record, including precise GPS geographic coordinates, device hardware identification, and a cryptographic date and time stamp, thereby creating a spatially and temporally verifiable chain of custody.

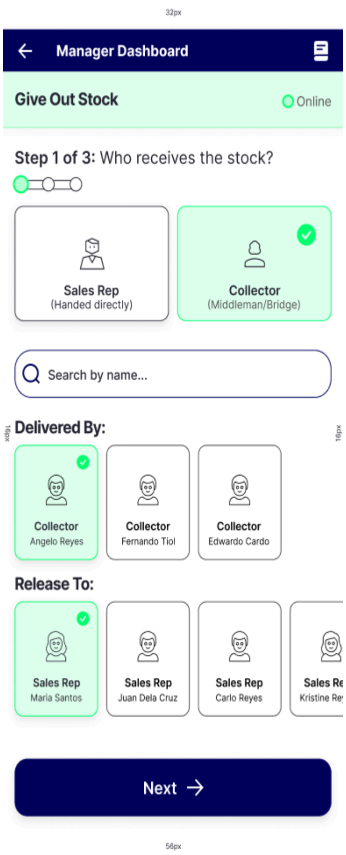


Figure 27

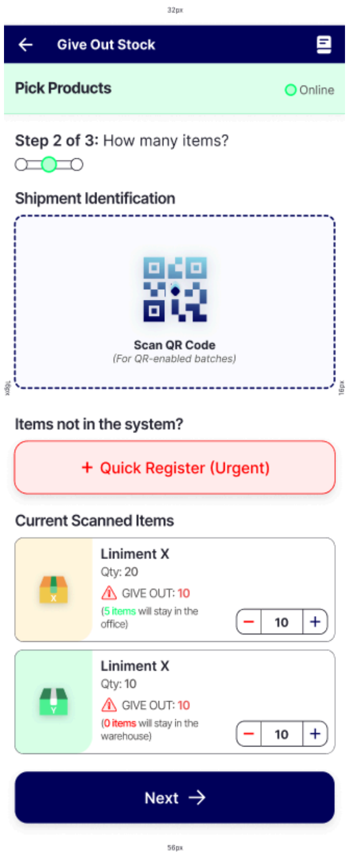


Figure 28

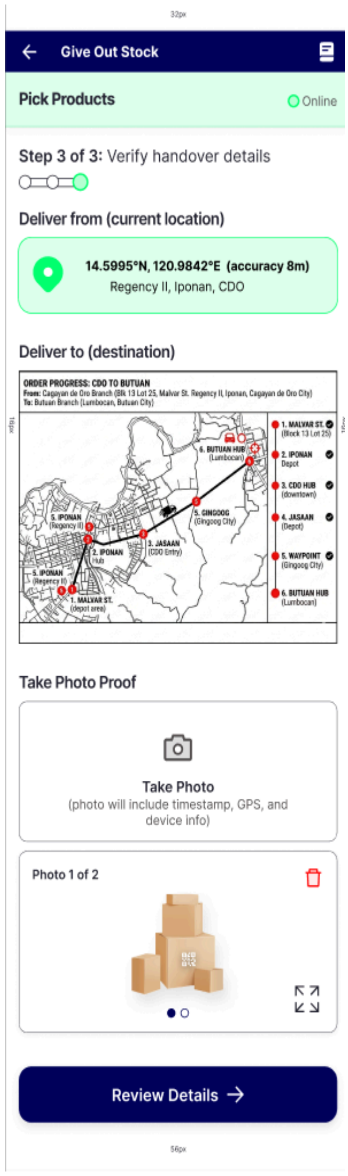


Figure 29

Figure 27 - 29. Manager Remote Stock Release for Sales Reps Interface

Figure 27 captures the interface for the distributed stock release module, representing the frontend framework for Process 5.0 (Stock Release Management). In contrast to the direct transaction channel depicted in Figure 24, this interface illustrates the remote distribution paradigm wherein a collector functions as an active courier intermediary. Within this workflow, the branch manager is equipped with structural selection mechanisms to simultaneously designate an authorized collector to facilitate logistics and select the target sales representative assigned to receive the product batch. The personnel records populating these respective selector components are dynamically populated via relational queries from the user_profiles_table, enforcing strict role validation and programmatic constraint checking prior to transaction initiation.

Figure 28 captures the second step interface for the remote stock release sequence. Through this interface, the branch manager can utilize the integrated mobile scanning hardware to decode pre-existing QR codes, dynamically retrieving the corresponding dataset from the branch_inventory_table. To ensure operational continuity during urgent periods where product batches lack pre-registered QR codes, the system incorporates a "Quick Register (Urgent)" protocol. This serves as an immediate navigational bridge back to the QR batch generation interface detailed in Figure 22, minimizing workflow friction. Once code verification is successfully executed, the system updates the session state, rendering the itemized product arrays within the "Current Scanned Items" buffer to allow for validation prior to ledger finalization.

Figure 29 captures the third step of the release stock via collector interface. Within this interface, the system integrates native geolocation hardware to determine and record the branch manager's current geographic position, establishing a verified point of origin for the inventory dispatch. Concurrently, the interface renders a dynamic "Deliver to (Destination)" map utility, which enables the administrative user to pinpoint and establish the target terminal location for the stock transfer. To supplement the geographic audit trail, a localized camera workflow is embedded to allow the manager to capture and attach real-time photographic evidence of the package or transit invoice, ensuring that both environmental coordinates and visual confirmation are logged before advancing to final verification.

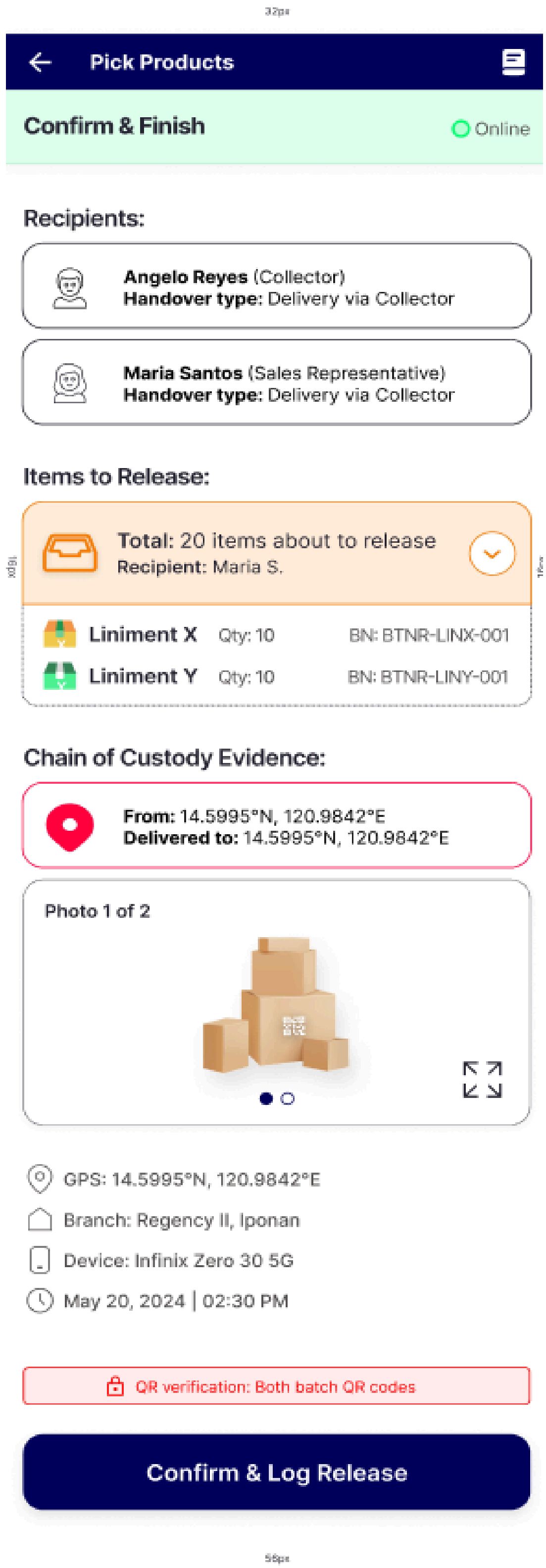


Figure 30. Manager Remote Stock Release Confirmation

Figure 30 captures the summary and confirmation interface for the release stock via collector. This interface aggregates the analytical data compiled from the preceding configuration phases, presenting the administrative user with an itemized breakdown of the scanned product arrays destined for the target sales representative. Crucially, the interface exposes the explicit spatial telemetry of the transaction, rendering the dual-point geographic coordinates that map the verified point of origin alongside the target terminal destination. Furthermore, it surfaces the unalterable metadata bound to the primary photographic proof container, specifically detailing the hardware-derived GPS coordinates, device specification, and localized timestamp attributes, thereby consolidating all evidentiary indicators into a single auditable record prior to executing the backend storage synchronization.

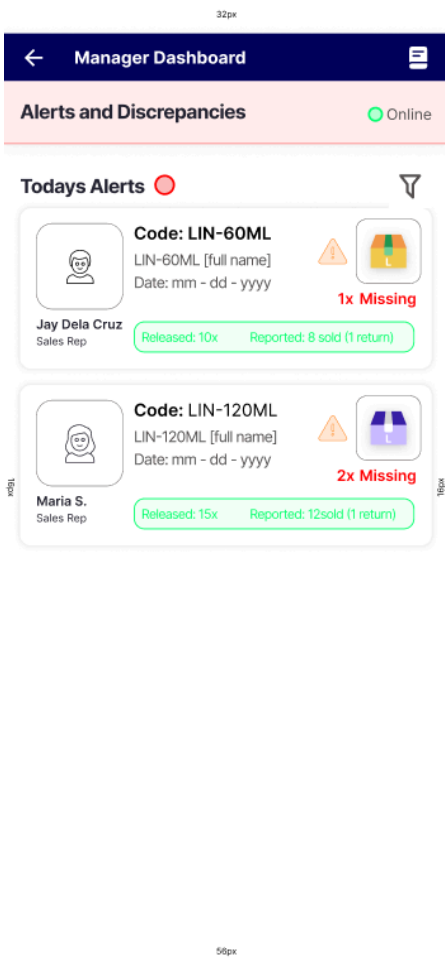


Figure 31. Manager Alerts Page

Figure 31 captures the manager's Alerts and Discrepancies Interface. This interface aggregates data compiled by the automated reconciliation engine, dynamically rendering an itemized overview of sales representatives currently exhibiting stock discrepancies relative to their records in the SR_inventory_table. For each flagged irregularity, the view exposes transaction details, providing the manager with complete operational visibility into the specific product code, full product name, and the amount of discrepancy detected. Crucially, the interface presents a comparative breakdown of key metrics, including the total quantity released, the total accounted stock units distributed between verified sales and returns, and the resulting calculation of missing inventory. This enables immediate root-cause investigation within the branch audit trail before discrepancies accumulate.

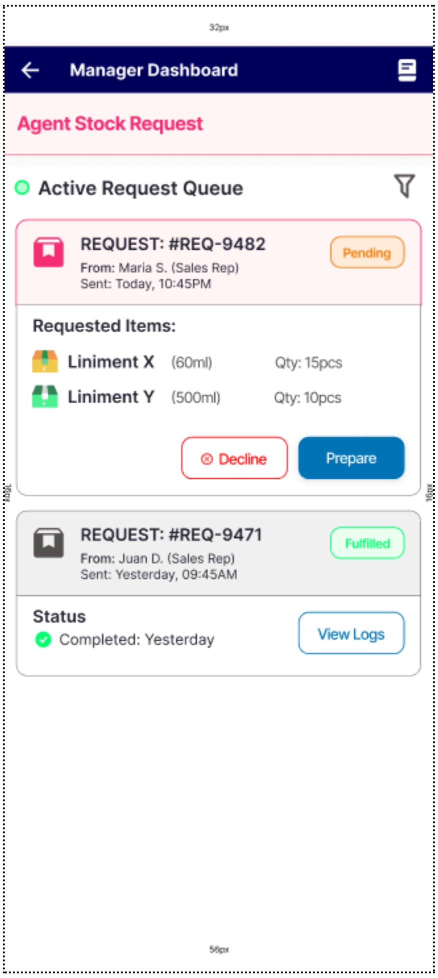


Figure 32. Manager Agent Requests Confirmation Interface

Figure 32 captures the request confirmation interface configured for the manager user tier, establishing an administrative fulfillment and gatekeeping layer for field-level inventory tracking. Within this interface, managers can evaluate incoming stock requests submitted by the sales representatives. Each request entry details critical tracking parameters, including the requester's identity, the submission timestamp, the target product line, and the requested distribution volume. Upon evaluating these metrics, the manager can execute two primary actions: "Prepare" to confirm and initiate the physical picking process, or "Decline" to reject the allocation request. Furthermore, a historical ledger is positioned below the active workspace, displaying resolved requests alongside their final administrative outcomes and providing a "Log" action node to review granular, deep-dive historical records.

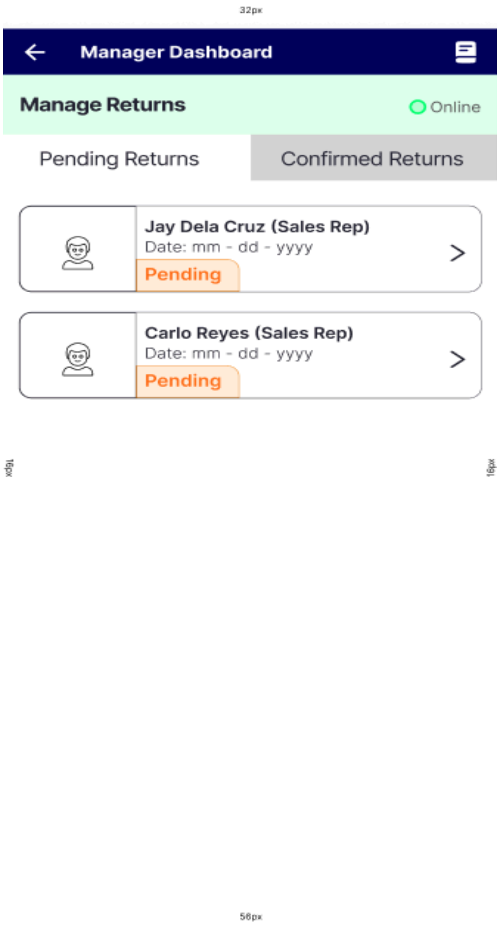


Figure 33

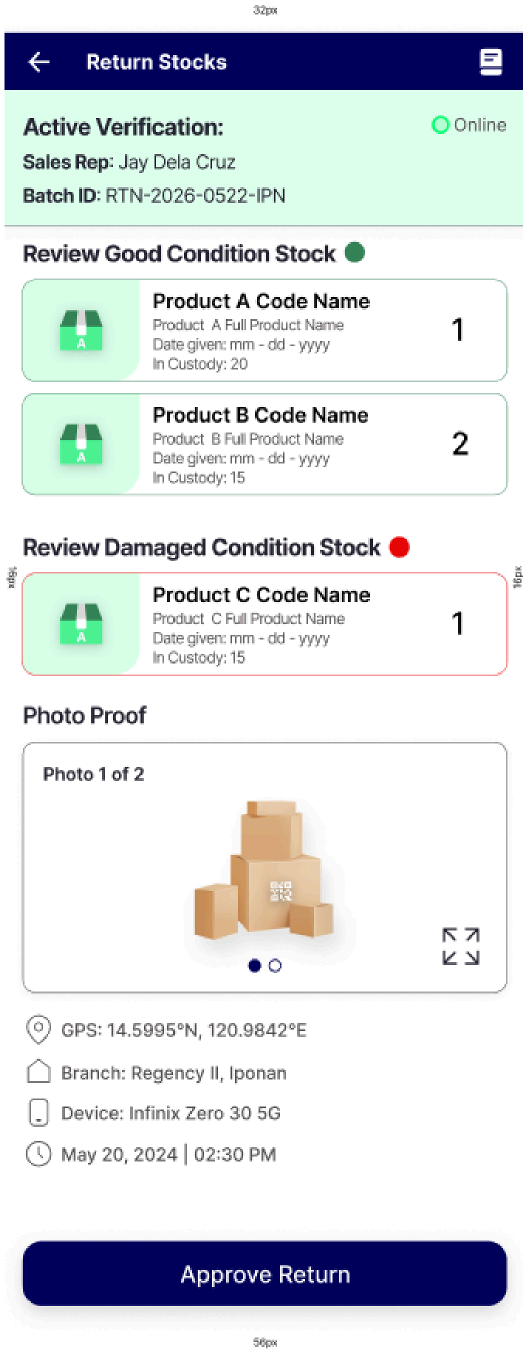


Figure 34

Figure 33 - 34. Manager Manage Return and Approval of Returns Interface

Figure 33 captures the return management interface configured for the manager user tier, establishing an administrative control layer for verification and inventory reconciliation. Within this workspace, managers are presented with a centralized queue of return requests submitted by decentralized sales representatives. The presentation layer features a dual-tabbed navigation architecture to segment workflows efficiently: the "Pending Approval" tab isolates active, unverified return payloads awaiting physical audit and ledger confirmation, while the "Confirmed Returns" tab surfaces a historical, read-only ledger of successfully reconciled and restocked inventory batches.

Figure 34 captures the itemized return verification interface configured for the manager user tier, serving as the final audit layer for field-level inventory reconciliation. Within this control view, managers are presented with a granular tracking ledger detailing the specific stock payload designated for return by an individual sales representative. The data matrix exposes critical parameters for each line item, including the exact quantity earmarked for return, the initial distribution timestamp, total units remaining in active field custody, and the physical condition classification. Additionally, for items marked as non-salable, the interface surfaces an embedded photographic evidence component containing verified device and environmental metadata, ensuring absolute accountability and a secure, immutable chain of custody before inventory restocking occurs.

Figure 35

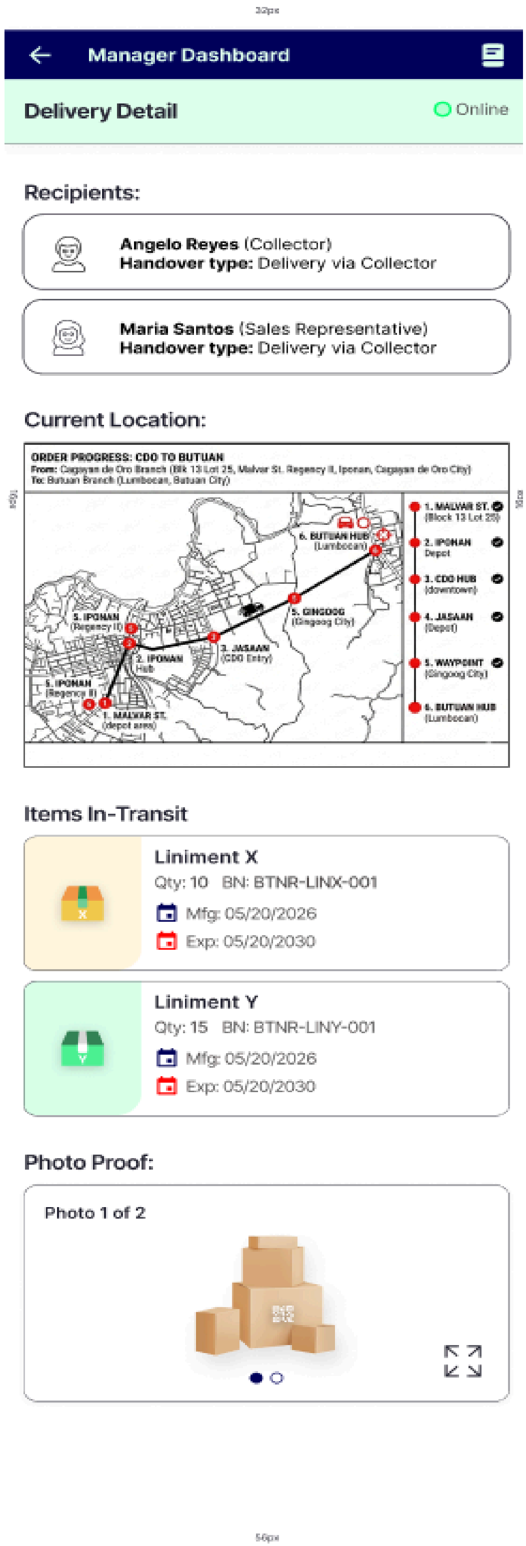


Figure 36

Figure 35 - 36. Manager Delivery Tracking Interface

Figure 35 captures the manager delivery tracking interface. This interface acts as a centralized monitoring hub, parsing active transit logs to render an overview of all collection agents currently deployed in the field for inventory delivery. By displaying the current operational distribution of the courier workforce, this interface serves as the primary administrative entry point for tracking remote logistics pipelines and reviewing ongoing transit events prior to field handover confirmation.

Figure 36 captures the delivery detail of the specific collector within the delivery tracking interface. This interface provides complete operational visibility regarding a specific collector delivery, aggregating the verified identity profiles of both the transporting collector and target sales representative recipients. Crucially, the dashboard renders a dynamic spatial overview tracking the collector's real-time geographic progression across designated transit checkpoints, alongside an itemized manifest of all product batches currently in transit. To preserve full evidentiary accountability, the screen surfaces the timestamped photographic proof captured at the point of origin, ensuring that the manager can systematically verify package condition, workforce telemetry, and stock allocation parameters from a single centralized interface.

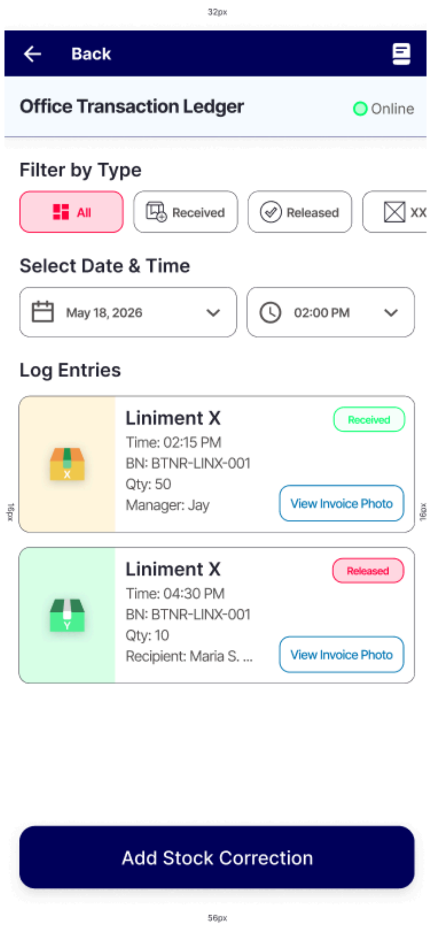


Figure 37. Manager Transaction Ledger Interface

Figure 37 captures the transaction ledger interface configured for the manager user tier, establishing a historical audit trail for end-to-end inventory tracking. Within this analytical workspace, the presentation layer surfaces an itemized record of all past logistical transactions, seamlessly indexing both stock receptions and distribution releases. Each historical log entry exposes critical data points, including automated transfer timestamps, unique batch identification numbers, precise transaction quantities, and designated recipient personnel profiles. Furthermore, the control layout embeds a dedicated “View Photo Proof” action node for each entry, allowing managers to immediately verify the physical condition manifest and maintain a secure, immutable chain of custody.

Figure 38. Manager Monthly Stock Report Summary Interface

Figure 38 captures the monthly stock report summary interface configured for the manager user tier, establishing a centralized data-aggregation layer for end-of-cycle inventory valuation. Within this analytical dashboard, managers are presented with an up-to-date monthly ledger detailing comprehensive inventory tracking parameters. The presentation matrix exposes key metrics for each stock line, including actual inventory volume received within the monthly cycle, consolidated sales figures reported by the sales representatives, returned unsold stock volumes, and quantified unresolved stock losses. Furthermore, the control layer provides three distinct utility nodes to streamline reporting distribution: "Share to Printer" for direct local hardware spooling, "Download Clean PDF" to generate and export static digital documents, and "Share to Chat" to securely transmit the reporting payload via integrated third-party messaging protocols.

3.8.2 Sales Representative POV System Prototype

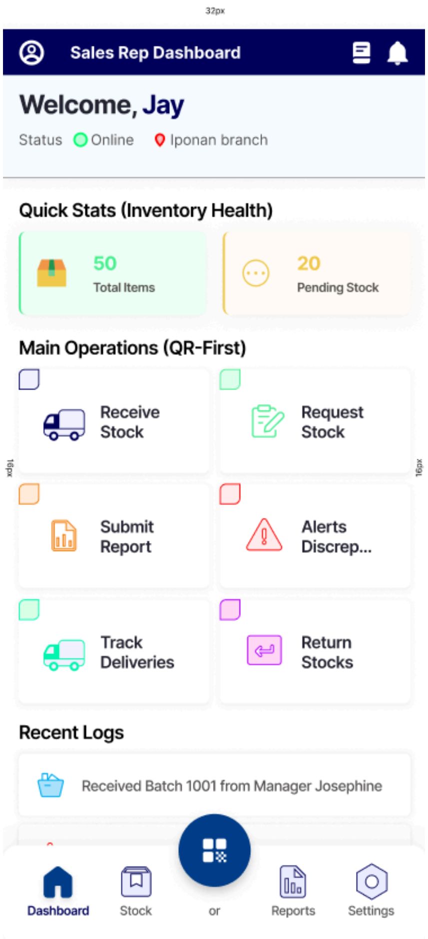


Figure 39. Sales Representative Dashboard

Figure 39 captures the primary application dashboard configured for the sales representative user tier, establishing the mobile frontend framework for decentralized field operations. Operating as an intuitive, role-specific gateway, this interface surfaces a centralized suite of transactional and reporting utilities designed to streamline field-level inventory workflows. The presentation layer displays high-level metric summaries, including current stock levels and pending stock requests, alongside rapid operational shortcuts. Through these dedicated navigation nodes, sales representatives can seamlessly initiate key tracking tasks, including receiving stocks in person via the “Receive Stock” button, request specific stock via the “Request Stock” button, daily reports via “Submit Report” button, discrepancy alert monitoring via "Alerts and Discrepancies" button, tracking stock deliveries via "Track Deliveries" button, and returning of stocks via "Return Stock" button.

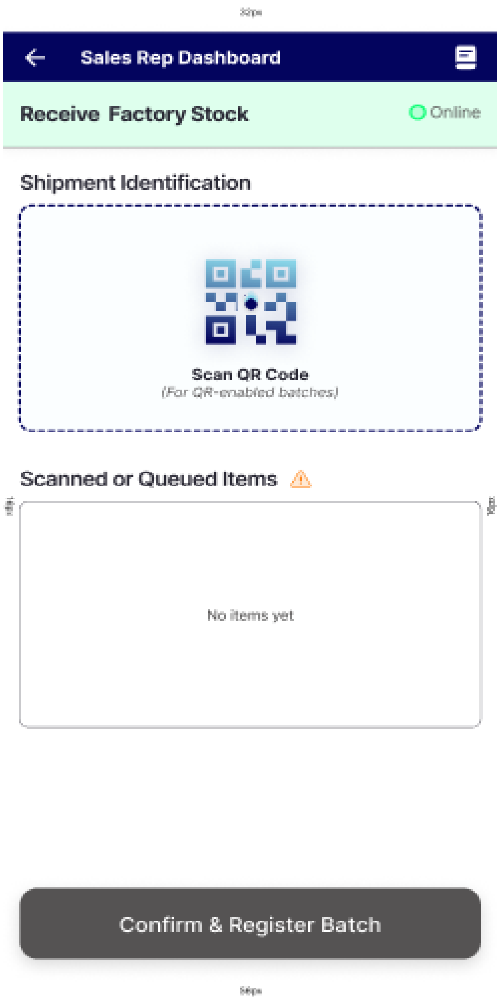


Figure 40

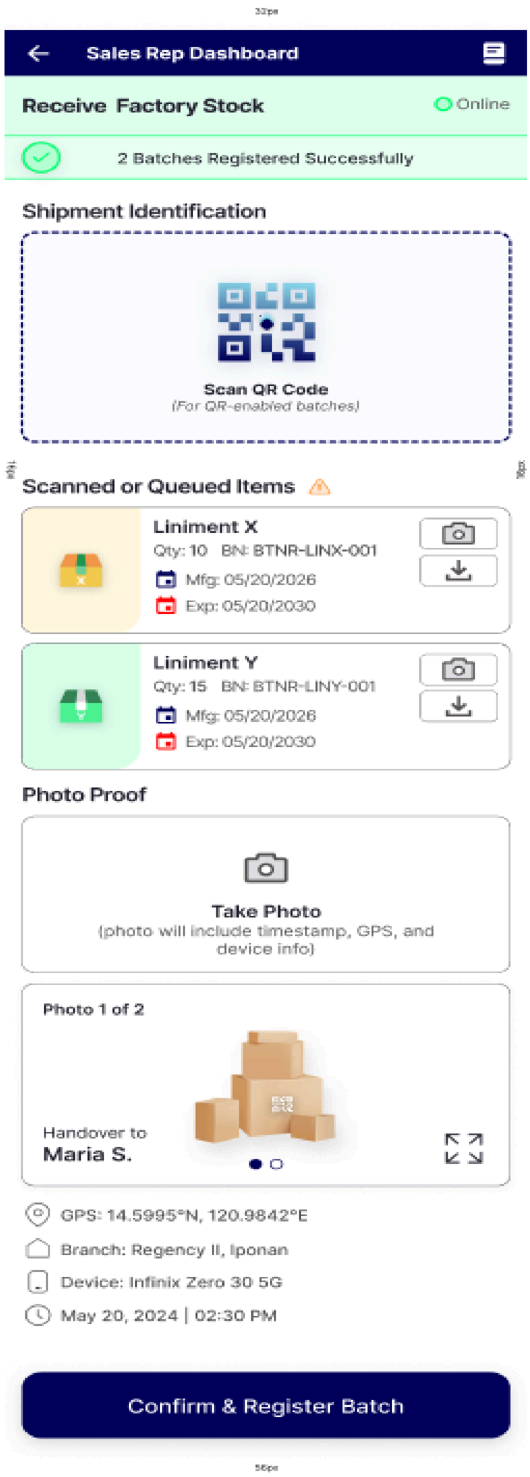


Figure 41

Figure 40 - 41. Sales Representative Receive Stocks Interface

Figure 40 captures the Sales Representative’s receive stock interface for the stock release sequence. Through this interface, the sales representative can utilize the integrated mobile scanning hardware to decode pre-existing QR codes. Once code verification is successfully executed, the system updates the session state, rendering the itemized product arrays within the "Current Scanned Items" buffer to allow for validation prior to ledger finalization.

Figure 41 captures the in-person stock receiving interface for the sales representative. Within this workflow, the user is required to scan the allocated product batch using the mobile device's QR code scanner. Once the scan is successfully executed, the system decodes the data and automatically displays the complete itemized list of products transferred by the branch manager. To prevent accountability disputes, the interface also enforces a mandatory photo confirmation step. This feature connects directly to the phone's native camera, blocking access to the photo gallery to ensure that only real-time pictures are accepted. Upon capturing the image, the system automatically saves the corresponding transaction metadata, including the precise GPS coordinates of the transfer location, the device model used, and the exact date and time.

Figure 42

Figure 43

Figure 42 - 43. Sales Representative Delivery Tracking Interface

Figure 42 captures the active delivery monitoring interface configured for the sales representative user tier. This screen parses the active transit ledger to render a list of all incoming shipments currently assigned to the logged-in agent account

Figure 43 captures the incoming shipment verification interface for the sales representative, which displays all parameters regarding an active courier delivery. This tracking screen consolidates the necessary transit data, identifying the designated delivery courier alongside the target representative account currently authorized to receive the shipment. Furthermore, the interface displays a tracking map showing the collector's current geographic location, an itemized list of all incoming product units, and the visual photo proof captured by the manager at the point of origin. Once the collector arrives at the field territory, the sales representative can click the "Scan and Receive Item" option to redirect them to figure 40, allowing them to scan the QR code and verify the batch items and log them directly into the SR_inventory_table.

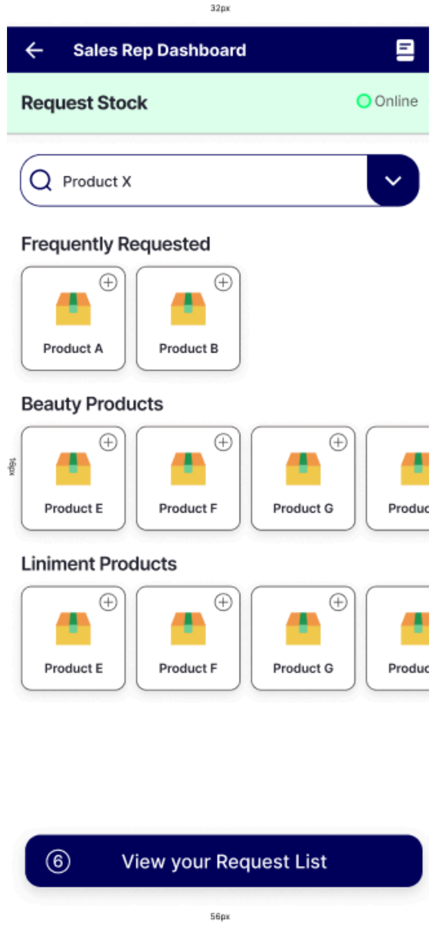


Figure 44

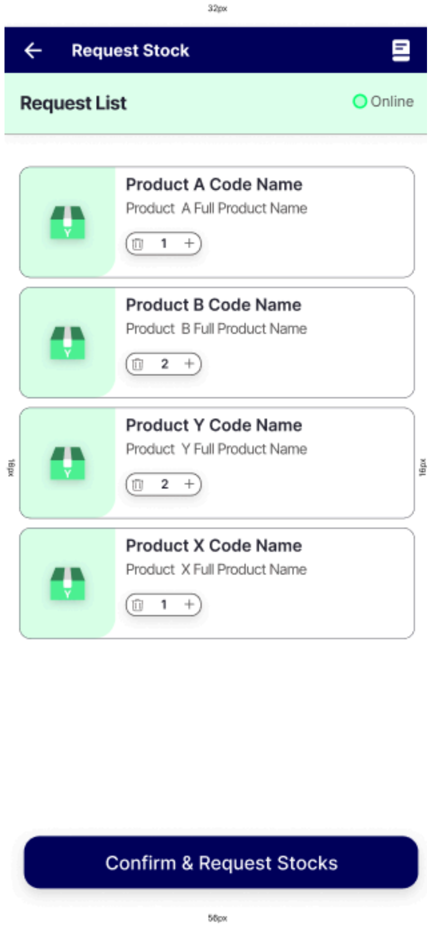


Figure 45

Figure 44 - 45. Sales Rep Stock Request Interface

Figure 44 captures the request stock interface configured for the sales representative user tier. Within this interface, sales representatives can browse through categorized product lines, such as beauty or liniment products, and select desired quantities of any inventory items they need to fulfill customer demand in their territory. Once a product count is adjusted and added, the application updates the session queue to track the selections. These queued items are then dynamically reflected within the "Request List" overview screen shown in Figure 45 for final validation.

Figure 45 captures the request list overview, which serves as the final step of the stock request workflow. Within this confirmation screen, the sales representative is presented with a complete summary of all their selected products prior to submission. To ensure a smooth user experience and eliminate unnecessary navigation back and forth between screens, interactive counter buttons are built directly into each item row. This design allows representatives to easily adjust item quantities up or down right from the summary list before tapping the confirmation button to send the finalized request to the branch manager.

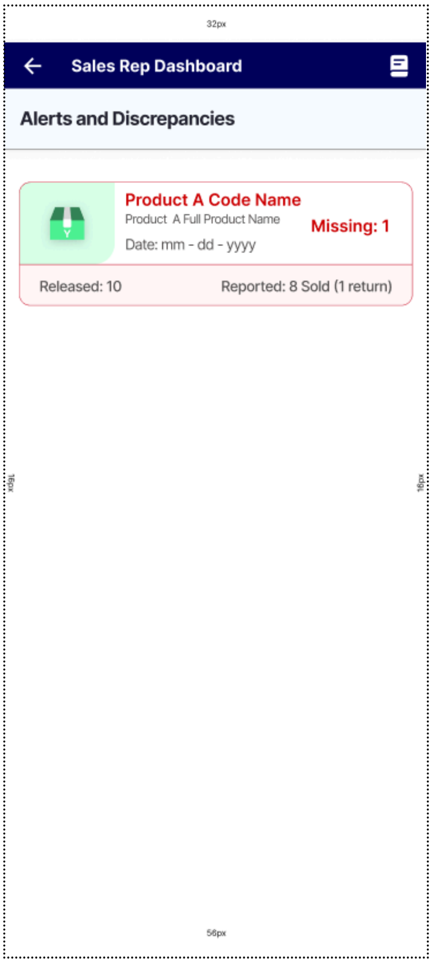


Figure 46

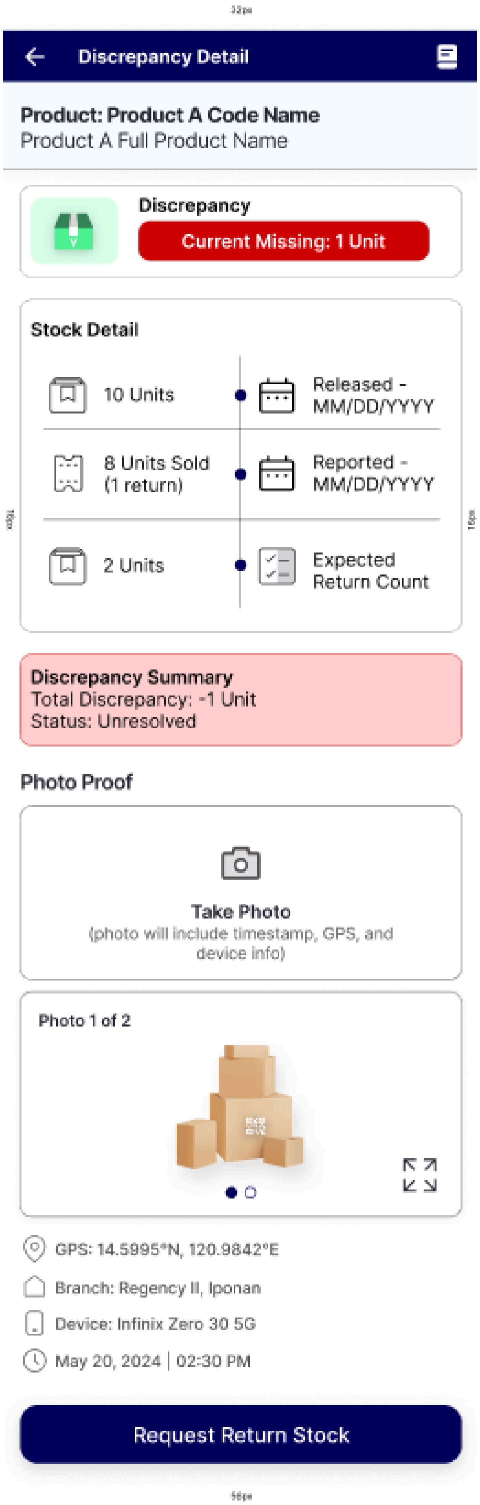


Figure 47

Figure 46 - 47. SR Stock Alert Detail and Discrepancy Resolution Interface

Figure 46 captures the alerts and discrepancies interface configured for the sales representative user tier. Within this interface, sales representatives are automatically notified following the submission of their daily reports if the automated reconciliation engine detects any variances in their stock levels. For any flagged mismatch, the interface displays granular transaction data, including the product code name, full product nomenclature, and the precise date of the alert. It also provides a clear comparative breakdown showing the total number of units initially released to the agent alongside the combined totals of what was reported as sold and returned. The system uses this comparison to calculate and explicitly display the exact number of missing units.

Figure 47 captures the discrepancy detail interface configured for the sales representative user tier, providing granular tracking visibility to resolve inventory discrepancies. Within this interface, sales representatives are presented with comprehensive ledger metrics surrounding the flagged discrepancy, including original inventory release volumes, precise distribution timestamps, and consolidated summaries of completed sales transactions and customer returns. The interface explicitly contrasts the system's expected inventory count against actual field data to isolate the variance. Furthermore, this control layer facilitates end-to-end discrepancy resolution by enabling agents to attach photographic evidence of the missing stock directly within the workflow before executing the reconciliation process via the dedicated "Request Return Stock" action node.

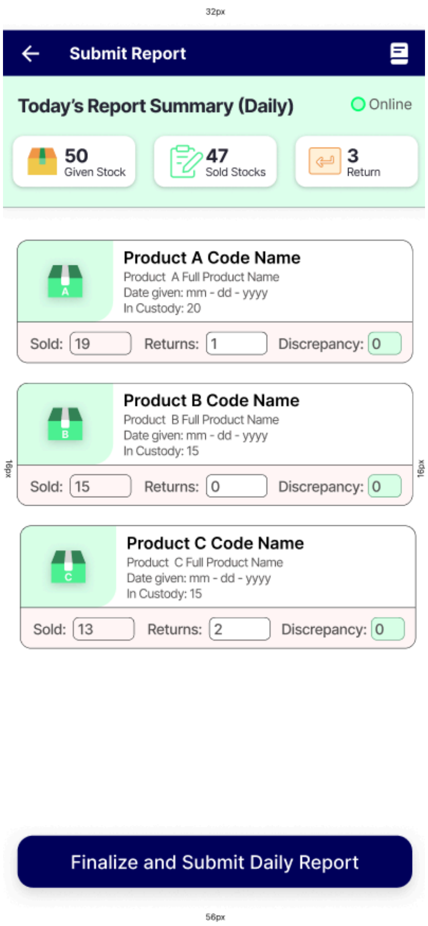


Figure 48. Sales Representative Submit Report Interface

Figure 48 captures the report submission interface configured for the sales representative user tier, establishing a structured reconciliation layer for daily field operations. Within this interface, sales representatives can compile and transmit a comprehensive daily inventory report based on the specific stock allocations received for that operational cycle. The interface dynamically displays items currently held in the agent's custody, providing interactive input fields to record precise units sold and items earmarked for return as unsold inventory. Based on these inputs, the system executes real-time balancing logic to detect and isolate quantity discrepancies before final submission. Upon engaging the “Finalize and Submit Daily Report” action node, the validated ledger payload is automatically routed to the branch manager's administrative interface for review and formal verification.

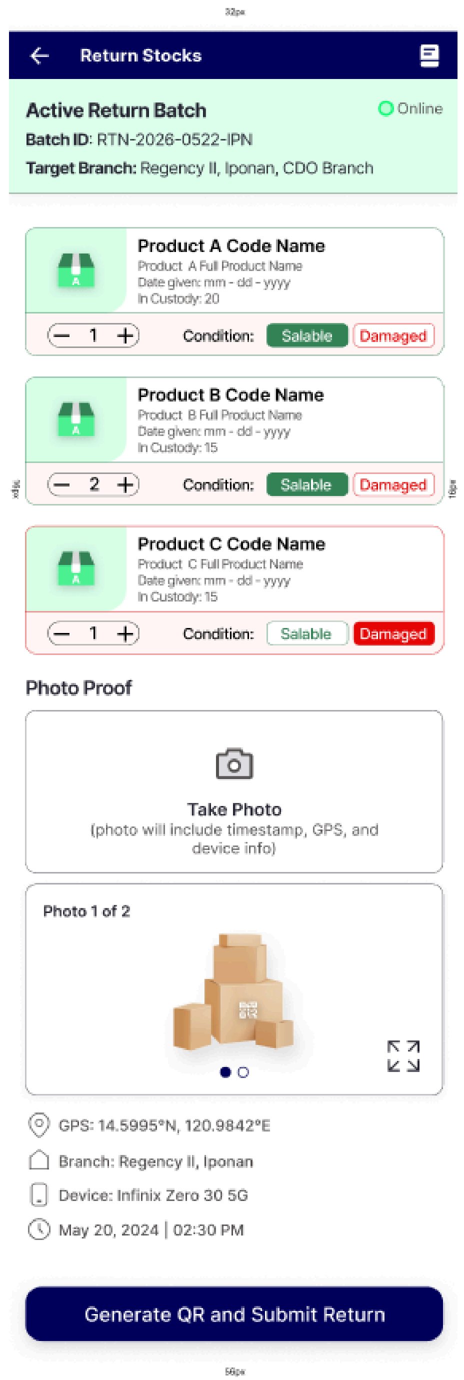


Figure 49. SR Return Stocks Interface

Figure 49 captures the return stocks interface configured for the sales representative user tier, providing a decentralized mechanism for reverse logistics and chain-of-custody tracking. Within this interface, sales representatives are presented with an itemized ledger of inventory currently in their possession. Each line item features interactive increment and decrement controls to specify the precise quantities slated for return. To ensure proper routing and financial accounting, the interface enables agents to categorize inventory by physical condition, explicitly separating pristine, "salable" units from "damaged/defective" stock. For items designated as damaged, the system enforces a mandatory photographic validation workflow that cryptographically embeds localized metadata, including precise GPS coordinates, device identifiers, and tamper-resistant timestamps, directly into the payload. Upon initiating the "Generate Return Manifest & QR Code" action node, a localized delivery manifest is rendered as a scannable QR code on the device while simultaneously transmitting a verification request to the branch manager's dashboard.

3.8.3 Collector POV System Prototype

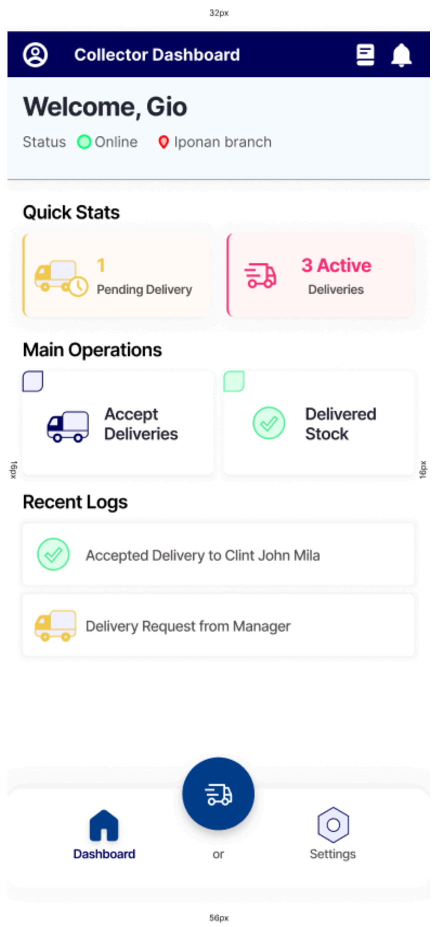


Figure 50. Collector Dashboard

Figure 50 captures the primary application dashboard configured for the collector user tier, establishing the operational frontend for logistical fulfillment and localized distribution tracking. Within this interface, users are presented with aggregate metrics summarizing active and pending delivery workloads alongside rapid operational shortcuts. The interface surfaces a targeted suite of execution nodes designed to manage the distribution lifecycle: the “Accept Deliveries” node facilitates the formal review and custody transfer of incoming warehouse dispatches; the “Delivered Stocks” utility provides a historical ledger of completed transmittals; and the prominent, centrally positioned action node enables agents to instantly initiate active routing and transit tracking workflows in the field.

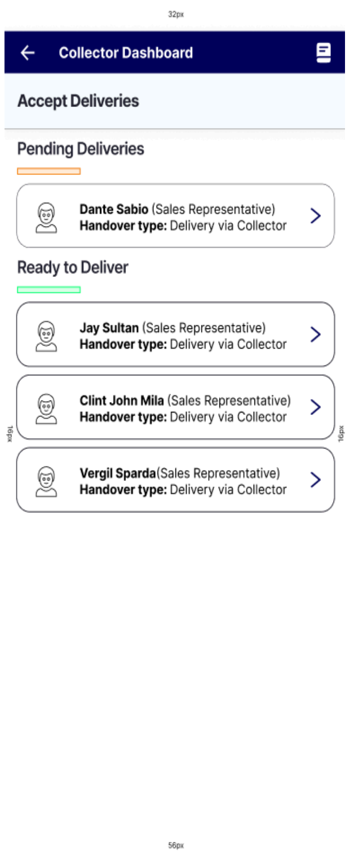


Figure 51

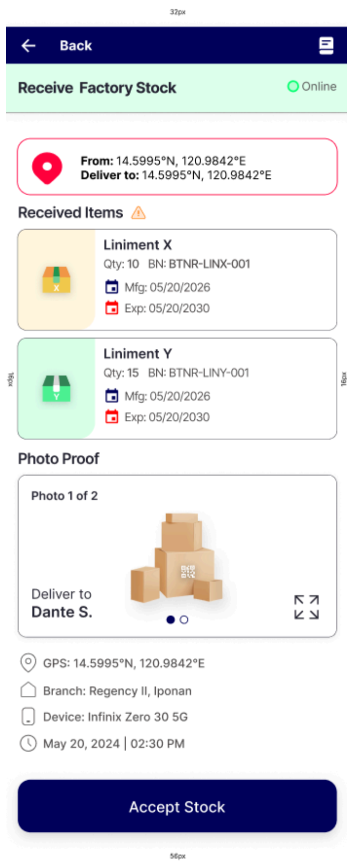


Figure 52

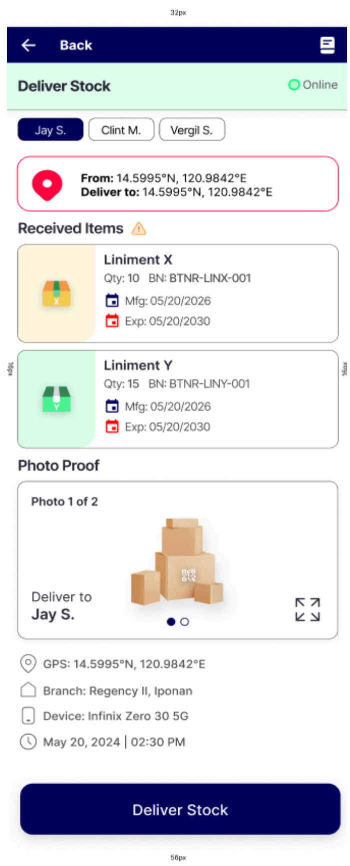


Figure 53

Figure 51 - 53. Collector Accept Deliveries and View Details Interface

Figure 51 captures the delivery confirmation interface configured for the collector user tier, establishing the operational checkpoint for localized logistics and fulfillment tracking. Within this interface, collectors are presented with pending distribution assignments dispatched from the manager's administrative console. Individual ledger cells expose critical handover parameters, explicitly displaying the target sales representative's identity alongside the designated custody transfer modality, which, within this workflow, is standardized as “Delivery via Collector.” The interactive presentation layer permits users to tap any specific record cell to expand its transactional details. A comprehensive structural breakdown of this sequence is illustrated sequentially in Figure 52.

Figure 52 captures the expanded delivery detail interface activated upon selecting a specific queue item. The interface presents a multifaceted logistical dataset to guide the user, exposing precise routing coordinates (differentiating the source pickup location from the target field delivery destination), an itemized manifest of the assigned stock batch payload, and the manager’s verified photoproof complete with embedded security metadata. Once the collector confirms delivery and assumes physical custody of the stocks, they execute the “Accept Stock” action node to officially log the chain-of-custody transfer within the system architecture.

Figure 53 captures the identical detailed delivery view structure as Figure 52, adapted exclusively for items categorized within the “Ready to Deliver” operational queue. This interface state represents a validated tracking phase where the logistics agent has already assumed physical custody of the stocks. Once the collector initiates the physical transport phase to the field destination, engaging the “Deliver Stock” action node dynamically activates the live transit and routing interface layer.

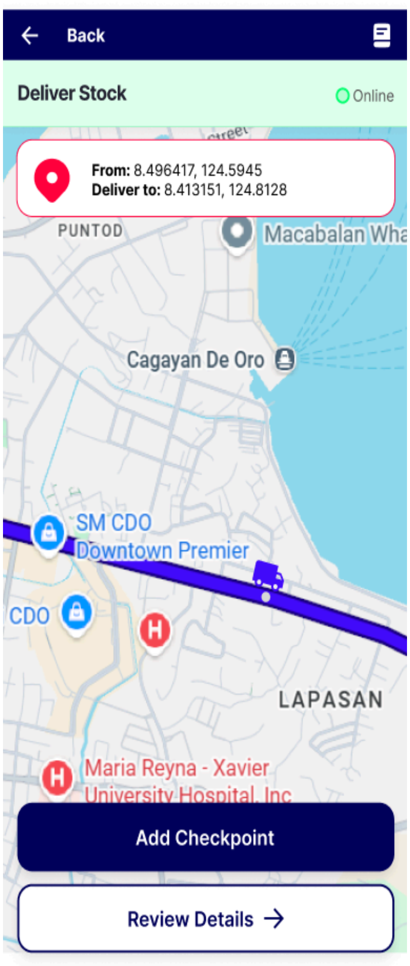


Figure 54

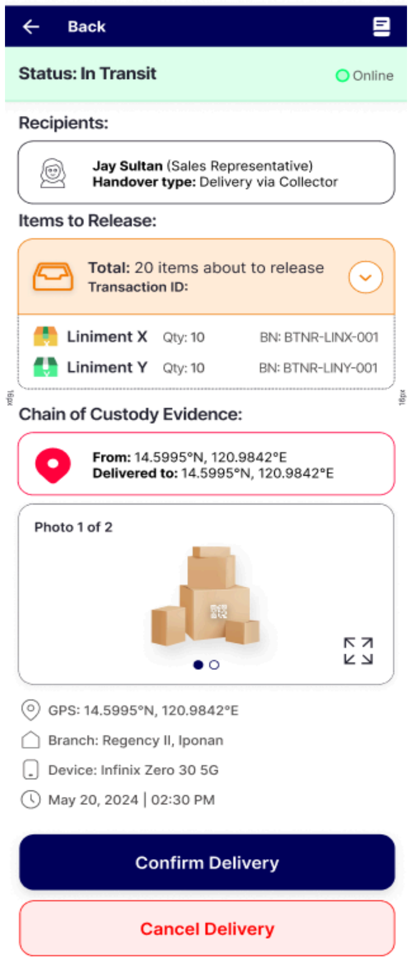


Figure 55

Figure 54 - 55. Collector Deliver Stock Process Interface

Figure 54 captures the live transit and routing interface configured for the collector user tier, serving as the active telemetry and geospatial tracking layer during transport operations. Within this interface, the application renders a dynamic map layer mapping real-time navigation paths from the collector's current coordinates to the target distribution destination. The core functional element is the "Add Checkpoint" action node; triggering this action executes an automated write operation that commits the collector's exact latitude and longitude coordinates directly to the gps_coord_table. This centralized data stream provides both branch managers and receiving sales representatives with transactional visibility to track transit intervals via sequential checkpoint logs.

Figure 55 captures the detailed delivery information interface structured for the active transport phase. While utilizing the same foundational layout architecture as Figure 52, this interface state exposes context-specific operational exceptions and resolution nodes tailored for in-transit tracking. Within this workspace, the collector is equipped with two primary state-resolution actions: “Cancel Delivery” to abort the transit assignment, or “Confirm Delivery” to formally close the workflow upon a successful handover and verify that the stock payload has arrived intact at its destination.

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